

Language-Guided Analogical Reasoning for Visible-Infrared Object Detection

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Abstract

Current visible-infrared (RGB-IR) object detection methods primarily rely on spatially aligned feature fusion, often neglecting intra-modal semantic similarities and inter-modal positional correspondences. As a result, detection performance has plateaued, particularly in complex backgrounds and small-object scenarios. In this work, we propose a novel *Language-Guided Analogical Reasoning for Visible-Infrared object Detection (GRID)*. The core idea is to leverage language representations as bridges that connect objects of the same category but with visually dissimilar features and uses visual similarity connections of one modality to construct object relations in another modality, ultimately achieving the transfer of useful features. Specifically, *GRID* includes two key modules: Language-Guided Objectness Attention (LGOA) and Semantic-Spatial Analogical Reasoning (S2AR). LGOA progressively enhances class text embeddings with RGB and IR region features and produces modality-specific objectness attention maps. S2AR constructs a graph with class text embeddings and selected RGB-IR pixel features as nodes, establishes semantic and spatial relation mappings, and propagates complementary information via two stages of graph reasoning, enabling challenging object detection. Extensive experiments on the DroneVehicle, M3FD, and

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LLVIP datasets demonstrate that **GRID** achieves state-of-the-art performance. The code, data, and models will be publicly released upon publication.

Keywords: Analogical Reasoning, Visible-Infrared Object Detection, Vision-Language Model

1. Introduction

Object detection, a pivotal task in computer vision, supports critical applications such as autonomous driving and intelligent traffic surveillance [1]. Conventional approaches, reliant on visible (RGB) imagery, leverage detailed texture and color information under optimal lighting conditions. However, their performance deteriorates significantly in challenging environments, including low-light settings, fog, or heavy rain. Conversely, infrared (IR) imagery provides consistent thermal signatures and clear object boundaries across diverse illumination and weather conditions. Nonetheless, the lack of color and fine texture in IR data limits its ability to differentiate visually similar objects [2, 3]. To address these constraints, multimodal object detection has emerged as an effective paradigm, harnessing complementary RGB and IR features to achieve superior accuracy and robustness in complex scenarios [4, 5].

Multimodal object detection methods can be classified into three types based on the fusion stage: *early fusion*, *middle fusion*, and *late fusion*. *Early fusion* adopts a simple single-branch architecture, combining RGB and IR images directly at the pixel level to form a unified input [6]. *Middle fusion* utilizes a dual-branch network to extract modality-specific features, followed by feature-level interactions through mechanisms such as inter-modal cross-attention [7], selective cross-modal fusion [8], target-aware attention [9], or uncertainty-aware cross-modal learning [10]. *Late fusion* integrates the outputs of independent modality-specific detectors, often using dynamic reweighting based on lighting conditions [11] or non-learning probabilistic ensemble techniques [12]. Among these, the middle feature-level fusion has significantly advanced RGB-IR object detection. However, these approaches typically rely on spatially aligned feature fusion, often overlooking intra-modal semantic similarities and cross-modal positional correspondences, limiting their ability to fully exploit multimodal

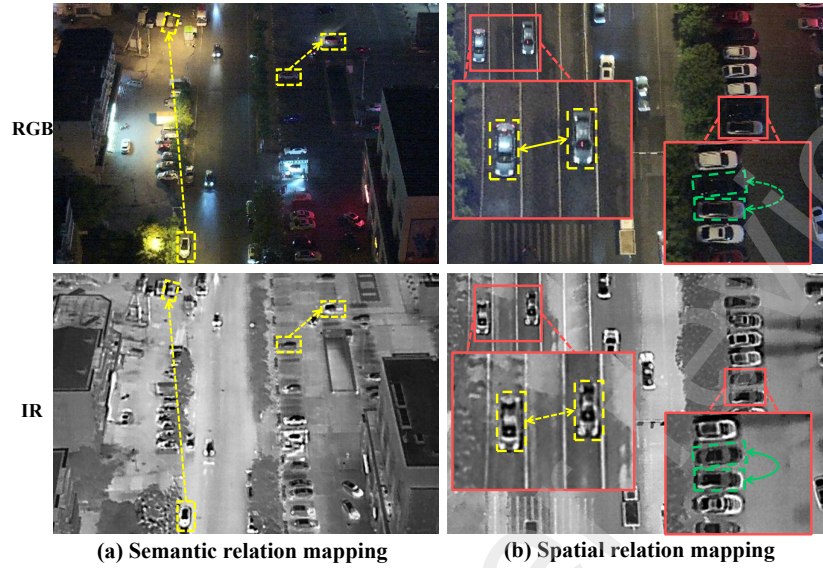


Figure 1: (a) Intra-modal semantic relation mapping: Semantic similarity among instances in terms of object-context sense. (b) Inter-modal spatial relation mapping: Vehicles in yellow dashed boxes show RGB similarity but differ in IR due to thermal variations; vehicles in green dashed boxes show IR similarity but differ in RGB due to color and lighting. Solid lines indicate actual visual similarity; dashed lines denote semantic similarity inferred through cross-modal relation mapping.

information.

Conventional fusion strategies in multimodal object detection struggle with domain discrepancies between RGB and IR modalities, arising from distinct imaging mechanisms that yield dissimilar visual features across spatial locations. These disparities hinder direct similarity computation, impeding robust cross-modal feature integration. Analogical reasoning, inspired by its success in knowledge graph reasoning [13] and visual analogy tasks [14], offers a promising approach by modeling object-level relation mapping. As illustrated in Fig.1(a), intra-modal semantic relations allow a region associated with the text embedding “a car on road” to guide the identification of similar objects despite visual differences. Fig.1(b) further demonstrates inter-modal spatial correspondences, class identical objects exhibit both similarity and divergence in different modalities. So, it is reasonable to construct a graph that integrates category-level

text features with RGB and IR pixel-level features, where text embeddings enriched with regional visual context serve as bridges to align dissimilar visual nodes; meanwhile, object similarity connection in one modality helps semantic information propagation across positions in another modality.

Based on the above analysis, we propose a novel method called *Language-Guided Analogical Reasoning for Visible-Infrared Object Detection (GRID)*. **GRID** comprises two key components: Language-Guided Objectness Attention (LGOA) and Semantic-Spatial Analogical Reasoning (S2AR). LGOA module harnesses CLIP’s semantic priors [15] to perform progressive cross-modal text enhancement with region visual features, generating refined textual embeddings that guide modality-specific objectness attention maps to mitigate representation gaps. S2AR module advances multimodal fusion through a two-stage analogical reasoning process: intra-modal analogical reasoning first enhances similar visual node features for each modality, followed by inter-modal analogical reasoning to integrate complementary contextual cues across modalities. This process follows a three-step pipeline [13]: abduction, constructing relation graphs from text and visual features; mapping, computing graph weights to map the semantic context and spatial correspondences; and induction, sequentially updating the graph to propagate complementary information.

Our contributions can be summarized as follows: (1) We introduce **GRID**, a novel language-guided analogical reasoning framework for RGB-IR object detection that refines modality-specific features and integrates cross-modal contextual cues, facilitating the detection of challenging objects by leveraging easier-to-detect instances. (2) A progressive text embedding enhancement method is proposed. This method can better integrate multi-scale regional visual features. These text features steer modality-specific objectness attention maps for RGB and IR, effectively mitigating representation disparities. (3) Semantic-spatial analogical reasoning advances feature fusion in a general sense. Specifically, intra-modal analogical reasoning refines RGB and IR pixel features based on contextual semantic similarity, while inter-modal analogical reasoning enables feature transfer across semantic regions.

2. Related works

2.1. Visible-infrared object detection

Multimodal object detection approaches for visible (RGB) and infrared (IR) data are broadly classified into early, middle, and late fusion strategies. Early fusion integrates RGB and IR images at the pixel level, aiming to directly exploit complementary low-level cues. For example, CSAA [6] adopts adaptive local-global learning to exploit complementary modality information. Middle fusion focuses on merging modality-specific deep features to achieve more flexible and robust representations. For instance, ICAFusion [16] introduces dual cross-attention transformers to simultaneously model global feature interactions and capture complementary information across modalities; DMM [9] leverages disparity-guided Mamba blocks to mitigate modality discrepancies; and ADMPF [17] incorporates a multibranch progressive fusion mechanism tailored for aerial imagery. Late fusion strategies operate at the decision level by combining predictions from individual modalities, thereby improving robustness. For example, MBNet [11] extends illumination-aware reweighting to the feature level, while ProbEn [12] employs a Bayesian probabilistic ensemble based on conditional independence assumptions. Despite these advances, reliance on spatially aligned feature fusion limits the exploitation of intra-modal semantic relationships and cross-modal positional correspondences, constraining performance in complex scenarios.

2.2. Analogical reasoning for object detection

Analogical reasoning facilitates object detection by inferring structural correspondences between entities under diverse contexts, allowing models to move beyond isolated object recognition toward relational understanding [18]. Existing approaches largely fall into two types: context-based reasoning and cross-domain transfer. The former exploits intra-image semantics, with C-GCN [19] modeling spatial-semantic relationships to enhance small object detection and RRNet [20] constructing region-based graphs to capture object interactions. The latter transfers external knowledge to novel categories, such as GRAN [21], utilizes vision-language graphs to facilitate zero-shot detection by bridging known and novel categories. However, these methods

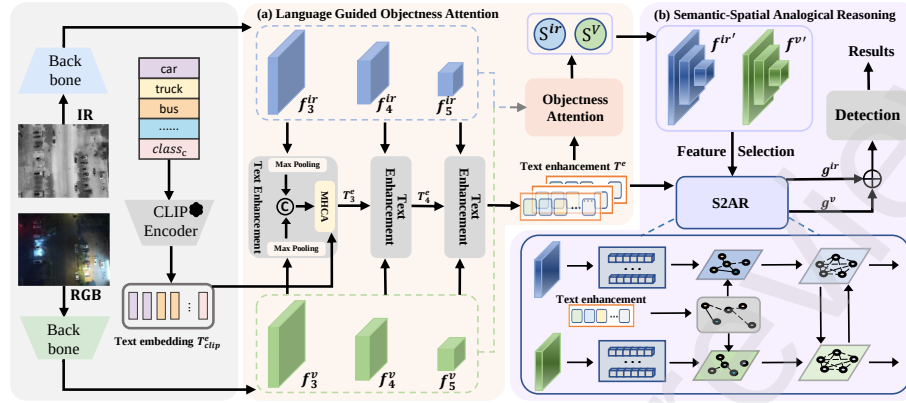


Figure 2: Overall framework of the proposed GRID. (a) The Language-Guided Objectness Attention module consists of progressive cross-modal text enhancement and semantic objectness attention, producing enhanced textual features, refined RGB and IR features, and objectness attention maps. (b) The Semantic-Spatial Analogical Reasoning module constructs relation graphs using text features and RGB/IR regions selected by objectness scores, models semantic and spatial correspondences, and applies two-stage graph reasoning to enhance feature representation.

primarily focus on generic relational reasoning without establishing structured mappings between object relations. SPAR [14] introduces self-prompted reasoning via CLIP in a single modality but lacks applicability to multimodal settings. In contrast, our proposed **GRID** extends relational reasoning by implementing cross-modal analogical relation mapping, enabling non-spatially aligned feature fusion to achieve robust RGB-IR object detection.

3. The Proposed Method

3.1. Problem statement

Let the training set $D = \{X^v, X^{ir}, Y\}$, where $X^v, X^{ir} = \{x_j^v, x_j^{ir}\}_{j=1}^{N_c}$ denote a collection of N_c paired visible (RGB) and infrared (IR) images, respectively. The set of labels $Y = \{y_j\}_{j=1}^{N_c}$ including class names and bounding boxes. $C = \{c_j\}_{j=1}^K$ is a set of classes that need to be detected, where c_j is the j -th class name and K is the number of classes. Our objective is to leverage language-guided analogical reasoning to enhance cross-modal feature representations, thereby improving RGB-IR object detection performance in

110 challenging scenarios.

Overview. Fig.2 illustrates the proposed *GRID* method, which comprises two components: Language Guided Objectness Attention (LGOA) module and Semantic-Spatial Analogical Reasoning (S2AR) module. Initially, a dual-stream backbone F_v and F_{ir} , based on the single-stage detector YOLOv8 [22], is adopted to extract multi-scale features f_i^v and f_i^{ir} , where $i \in \{3, 4, 5\}$, from the paired RGB and IR input images (x^v, x^{ir}) . LGOA module is designed to generate context-aware textual embeddings and objectness attention maps. Specifically, it employs a pooling-based attention mechanism to extract region-aware features from RGB and IR modalities, refining text embeddings into scale-adaptive T_i^e through progressive cross-modal integration. These enhanced embeddings are then fused with modality-specific features f_i^v and f_i^{ir} to generate objectness attention maps s_i^v and s_i^{ir} for RGB and IR, respectively. The resulting maps also reweight the corresponding features, yielding the updated features $f_i^{v'}$ and $f_i^{ir'}$, respectively. In this way, the visual features combine semantic information, enabling a correlation between visual features and text features.

125 Analogical reasoning in S2AR module unfolds in three steps: (1) Abduction, which constructs graphs at each feature level using category-level nodes from text embeddings T_i^e and pixel-level nodes independently selected from $f_i^{v'}$ and $f_i^{ir'}$ based on fused objectness scores; (2) Mapping, which establishes semantic relation mapping within each modality (text $\leftrightarrow$ RGB/IR) and spatial relation mapping between modalities (RGB $\leftrightarrow$ IR); (3) Induction, which updates node features in a sequential manner, first performing intra-modal semantic reasoning followed by inter-modal spatial reasoning. The updated node features are then projected back and fused to generate robust multimodal representations for final object detection.

3.2. Language Guided Objectness Attention

135 Effective fusion of RGB and IR features remains challenging due to significant discrepancies in their imaging properties and semantic granularity. In particular, the same object may present distinct appearance patterns across modalities and locations, making it difficult to establish meaningful similarity connections between them. Such inconsistency hinders the analogical reasoning required for robust multimodal object

140 detection. To address this, we bind CLIP cross-modal priors with contextual image features. This approach enhances textual embeddings with scene-specific semantics and imbues modality-specific feature maps with category-aware representations, facilitating robust semantic localization and mitigating pixel-level misalignments for effective multimodal fusion.

145 **Progressive cross-modal text enhancement.** To embed category-aware semantic priors into the visual representations and bridge the gap between RGB and IR modalities, we begin by constructing K category-specific prompts (e.g., “a photo of [category]”) based on a predefined label set $\{c_j\}_{j=1}^K$. These prompts are encoded using the CLIP text encoder [15], yielding the initial text embeddings $T^e \in \mathbb{R}^{K \times D}$, where D denotes the embedding dimension. For each feature level $i \in \{3, 4, 5\}$, we extract context-aware visual tokens from RGB and IR features f_i^v and f_i^{ir} by applying a 3×3 max pooling layer to capture local region representations, followed by flattening into a $9 \times D$ matrix:

$$\hat{f}_i^{v/ir} = \text{Flatten}(\text{MaxPool}_{3 \times 3}(f_i^{v/ir})), \quad (1)$$

where $\hat{f}_i^{v/ir} \in \mathbb{R}^{9 \times D}$. The pooled RGB and IR features are then concatenated to form a unified visual context $f_i = \text{Concat}(\hat{f}_i^v, \hat{f}_i^{ir})$ where $f_i \in \mathbb{R}^{18 \times D}$, which serves as a scale-aware and context-sensitive summary of the input features.

To progressively enrich the textual representations with multimodal visual context, we apply a Multi-Head Cross-Attention (MHCA) mechanism [23] to integrate the visual context f_i with text embeddings. At $i = 3$, the text embedding is computed as $T_3^e = T_{clip}^e + \zeta \Delta T_3^e$, where T_{clip}^e is the initial embedding from the CLIP text encoder. For $i = 4$, leveraging the refined embedding from $i = 3$, the embedding is $T_4^e = T_3^e + \zeta \Delta T_4^e$, and similarly at $i = 5$. Here, each ΔT_i^e is computed via the MHCA mechanism between the current text embedding and image features at scale i , with ζ as a hyper-parameter. This progressive refinement allows the text embeddings to incorporate hierarchical visual cues, adapting to the semantic variability of objects across scales and modalities.

165 **Remark.** The proposed progressive fusion strategy enables text embeddings to incrementally incorporate cross-modal visual context from different scales, enhancing their adaptability to scene semantics and discriminative ability. The resulting embed-

170 dings serve as dynamic semantic priors for guiding objectness-aware feature modulation and provide informative cues for subsequent semantic-spatial analogical reasoning. The effectiveness of this module is further validated in the Appendix.

Semantic objectness attention. Leveraging CLIP robust zero-shot generalization to identify semantically salient regions and suppress irrelevant background [24], we introduce a semantic objectness attention module that generates modality-specific attention maps guided by context-aware text embeddings T_i^e . We first compute class-aware 175 relevance maps $p_i^{v/ir}$ by measuring the similarity between pixel-level features $f_i^{v/ir}$ and refined text embeddings T_i^e :

$$p_i^{v/ir} = \beta \cdot \text{FC}_1(f_i^{v/ir}) \cdot \text{FC}_2(T_i^e) + \gamma, \quad (2)$$

where FC_1 and FC_2 are fully connected layers projecting visual and textual features into a shared embedding space, $\cdot$ denotes dot-product similarity, and β, γ are learnable 180 scalar parameters. The relevance map $p_i^{v/ir} \in \mathbb{R}^{H_i \times W_i \times K}$ category-specific likelihoods at each spatial location. Objectness attention maps are then derived via a maximum operation over the category dimension, followed by sigmoid activation:

$$s_i^{v/ir}(h, w) = \text{Sigmoid}\left(\max_k(p_i^{v/ir}(h, w, k))\right), \quad (3)$$

where (h, w) denotes the spatial coordinate, and $s_i^{v/ir} \in \mathbb{R}^{H_i \times W_i \times 1}$ indicates pixel-wise object confidence. The visual features are updated as follows:

$$f_i^{v/ir'} = f_i^{v/ir} \odot s_i^{v/ir} + f_i^{v/ir}, \quad (4)$$

185 where $\odot$ denotes element-wise multiplication. This modulation enhances likely object regions while preserving base feature integrity, mitigating spatial misalignments and modality-specific variations to provide robust representations for subsequent cross-modal analogical reasoning.

3.3. Semantic-Spatial Analogical Reasoning

190 Analogical reasoning achieves cross-position and cross-modal feature fusion. Firstly, we construct a graph that consists of class text feature nodes and RGB and IR pixel-level feature nodes. The class text feature nodes, due to their absorption of regional

visual features, can serve as a bridge for knowledge communication between pixel feature nodes that originally had dissimilar visual features. In practice, it is found that it is better to first perform the intra-modal analogical reasoning to enhance similar node features, and then perform inter-modal analogical reasoning fusion instead of full graph reasoning. For example, small or occluded cars are first inferred from similar car instances in other positions of the same modality. Then, in low-light conditions, IR thermal signatures of cars in other positions complement RGB features that lack structural details. The following statement follows standard analogy reasoning paradigm [13]: abduction, mapping, and induction.

Abduction: graph construction. In the initial phase of analogical reasoning, we construct multiple subgraphs at each feature level $i \in \{3, 4, 5\}$ to capture semantic and spatial relationships, utilizing category-level and pixel-level nodes. Category-level nodes are derived from refined text embeddings T_i^e , with each class $k \in \{1, \dots, K\}$ forming a node in the semantic node matrix $H_i^c \in \mathbb{R}^{K \times D}$. Pixel-level nodes are selected from enhanced RGB and IR features $f_i^{v'}$ and $f_i^{ir'}$, guided by fused objectness scores. An adaptive pixel selection strategy computes modality-specific weights via:

$$w_i^v, w_i^{ir} = \text{Softmax}([\mu(s_i^v), \mu(s_i^{ir})]), \quad (5)$$

where $\mu(\cdot)$ denotes the spatial mean, and $[\cdot, \cdot]$ indicates concatenation. The fused score map is:

$$s_i = w_i^v \cdot s_i^v + w_i^{ir} \cdot s_i^{ir}. \quad (6)$$

An adaptive threshold, $\tau_i = \mu(s_i) + \sigma(s_i)$, where $\sigma(\cdot)$ is the spatial standard deviation, identifies discriminative regions. Pixels with scores $s_i(h, w) > \tau_i$ are ranked, selecting the top- M positions $\{(h_1, w_1), \dots, (h_M, w_M)\}$, shared across modalities, to form pixel-level node matrices $H_{i,p}^v, H_{i,p}^{ir} \in \mathbb{R}^{M \times D}$.

Mapping: graph weight computation. (1) text-RGB/IR node weight matrix computation. The text-RGB adjacency matrix $\tilde{A}_i^{c:v} = [\tilde{A}_i^{c:v}(m, n)] \in \mathbb{R}^{N_v \times N_v}$, where $N_v = K + M$, can be computed as follows via pairwise similarity,

$$\begin{aligned} \tilde{A}_i^{c:v}(m, n) = & \text{Conv}_{1 \times 1}(H_i^{c:v}(m)) \cdot \tilde{\Lambda}(H_i^{c:v}) \\ & \cdot \text{Conv}_{1 \times 1}(H_i^{c:v}(n))^T, \end{aligned} \quad (7)$$

where the matrix $H_i^{c:v} = H_i^c \cup H_{i,p}^v$ integrates text and RGB pixel-level feature nodes. $\text{Conv}_{1 \times 1}$ denotes a 1×1 convolution with ReLU activation, and $\tilde{\Lambda}(H_i^{c:v}) \in \mathbb{R}^{D \times D}$ is a diagonal matrix learned to refine the distance metric:

$$\tilde{\Lambda}(H_i^{c:v}) = \text{diag}(\text{Conv}_{1 \times 1}(\text{AvgPool}(H_i^{c:v}))). \quad (8)$$

Similarly, we can get the text-IR adjacency matrix $\tilde{A}_i^{c:ir}$. These two matrices implement the mapping of the relations between the object and region context in different modalities.

(2) RGB/IR node weight matrix computation. For pixel-level node matrices H_i^v, H_i^{ir} , adjacency matrices $\tilde{A}_i^v$ and $\tilde{A}_i^{ir}$ are computed using Eq.(7) and Eq.(8). To enable adaptive cross-modal spatial relation reasoning, a dynamic gating mechanism fuses RGB and IR spatial structures. Graph features are transformed: $H_i^{v'} = \text{Conv}_{1 \times 1}(H_i^v)$ and $H_i^{ir'} = \text{Conv}_{1 \times 1}(H_i^{ir})$. The gate weights for RGB are

$$\alpha_i = \text{Softmax}(w_g([\text{AvgPool}(H_i^{v'}); \text{AvgPool}(H_i^{ir'})])), \quad (9)$$

where w_g is a 1×1 convolutional network, and $\alpha_i = [\alpha_i^v; \alpha_i^{ir}]$ adjusts modality contributions. Similarly, $\beta_i = [\beta_i^v; \beta_i^{ir}]$ is computed for IR by swapping inputs. The fused adjacency matrices are

$$\begin{bmatrix} \hat{A}_i^v \\ \hat{A}_i^{ir} \end{bmatrix} = \begin{bmatrix} \alpha_i^v & \alpha_i^{ir} \\ \beta_i^v & \beta_i^{ir} \end{bmatrix} \begin{bmatrix} \tilde{A}_i^v \\ \tilde{A}_i^{ir} \end{bmatrix}. \quad (10)$$

The above operation adaptively integrates cross-modal spatial dependencies and achieves complementarity of paths between modalities for subsequent reasoning.

Induction: two stages of graph reasoning. Graph convolution is used for reasoning by updating node features at each feature level based on the adjacency matrices defined in the mapping stage. We first perform graph reasoning on two subgraphs with nodes $H_i^c \cup H_{i,p}^v$ and $H_i^c \cup H_{i,p}^{ir}$, corresponding to the weight matrices $\tilde{A}_i^{c:v}$ and $\tilde{A}_i^{c:ir}$ respectively. The first stage reasoning uses text node features as bridges, propagates features from other locations to semantically similar pixel nodes, i.e., the corresponding node features will be updated. The second stage reasoning is performed on two subgraphs with nodes H_i^v and H_i^{ir} which the weight matrices $\hat{A}_i^v$ and $\hat{A}_i^{ir}$ respectively.

This stage reasoning, relying on cross-modal edge matrices, propagates semantic features from other locations to pixel nodes that are similar in other modality. Specifically, all of these reasoning can be computed as

$$H_i^\phi = \text{ReLU}(L_i^\phi H_i^\phi W_i^\phi), \quad (11)$$

where $\phi \in \{c : v, c : ir, v, ir\}$, H_i^ϕ represents the node matrix, $W_i^\phi \in \mathbb{R}^{D \times D}$ is a learnable weight matrix, and the normalized graph Laplacian matrix $L_i^\phi = I - \tilde{D}^{-1/2} \tilde{A}_i^\phi \tilde{D}^{-1/2}$ for $\phi \in \{c : v, c : ir\}$ and $L_i^\phi = I - \hat{D}^{-1/2} \hat{A}_i^\phi \hat{D}^{-1/2}$ for $\phi \in \{v, ir\}$.

Afterwards, the updated pixel-level node features H_i^v and H_i^{ir} are mapped back to the spatial positions $\{(h_1, w_1), \dots, (h_M, w_M)\}$ of the input feature maps $f_i^{v/ir'}$ respectively, yielding $\hat{g}_i^{v/ir} \in \mathbb{R}^{C \times H \times W}$. Then they are fused with $f_i^{v/ir'}$ via a residual connection as follows,

$$g_i^{v/ir} = \gamma_i^{v/ir} \cdot \hat{g}_i^{v/ir} + (1 - \gamma_i^{v/ir}) \cdot f_i^{v/ir'}, \quad (12)$$

where $\gamma_i^{v/ir}$ is an adaptive fusion weight computed from $\hat{g}_i^{v/ir}$ through global average pooling, a 1×1 convolution, and a sigmoid activation. The resulting features g_i^v and g_i^{ir} are aggregated element-wise to produce robust multimodal representations for detection.

Remark. Two-stage reasoning outperforms full graph reasoning working on all text, RGB, and IR nodes, which risks introducing noise from heterogeneous modalities and increases computational complexity. By decoupling semantic and spatial interactions, two-stage reasoning achieves robust feature enhancement and efficient cross-modal fusion, as further validated by the ablation studies presented later. Compared with previous pixel-level fusion methods, our method achieves cross-position semantic fusion based on semantic relation and spatial relation mappings.

Training and Inference. During training, our method employs a loss function aligned with the YOLO-World detector [25], defined as:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{con}} + \mathcal{L}_{\text{iou}} + \mathcal{L}_{\text{dff}}, \quad (13)$$

where $\mathcal{L}_{\text{con}}$ is a region-text contrastive loss, computed via cross-entropy between object-text similarity and their corresponding assignments, while $\mathcal{L}_{\text{iou}}$ (IoU loss) and $\mathcal{L}_{\text{dff}}$ (dis-

tribution focal loss) optimize bounding box regression. The parameters of CLIP text encoder remain fixed during both training and inference.

4. Experiments

270 4.1. Experimental setup

Datasets. We conduct comprehensive evaluations of our method on three public RGB-IR datasets. **DroneVehicle** [10] comprises 28,439 aligned RGB-IR image pairs captured from aerial perspectives under varied lighting conditions in urban and road scenes, annotated with five vehicle categories (car, bus, truck, van, freight car). The dataset is partitioned into 17,990 training, 1,469 validation, and 8,980 test pairs. We report performance on both validation and test sets. **M3FD** [26] contains 4,200 partially aligned RGB-IR image pairs from urban roads and campus scenes, covering six object categories (person, car, bus, truck, motorcycle, lamp). Following the scene-based partitioning in [27], we divide the dataset into 3,360 training and 840 test pairs. **LLVIP** [28] focuses on pedestrian detection under low-light conditions. It contains 15,488 aligned image pairs, with 12,025 for training and 3,463 for testing.

Implementation details. All experiments are conducted on an NVIDIA RTX 3090 GPU with 24 GB of memory, using an environment configured with CUDA 12.1 and PyTorch 2.4.1. We report detection results on the oriented bounding box (OBB) task using the DroneVehicle dataset, and on the horizontal bounding box (HBB) task using the M3FD and LLVIP datasets. All input images are uniformly resized to 640×640. Our method is developed based on a dual-stream YOLO architecture [29], extended with modules for intra-modal and inter-modal reasoning. Model optimization is performed using stochastic gradient descent (SGD) with a momentum of 0.937 and a weight decay of 0.0005. The learning rate is initialized at 0.01 and linearly decays to 0.0001 during training. The number of training epochs is set according to each dataset: 60 epochs for DroneVehicle, 180 for M3FD, and 50 for LLVIP. For the OBB task on DroneVehicle, we follow the standard evaluation protocols [30, 31], initializing **GRID** with YOLOv8-OBB pretrained models. For the HBB tasks on M3FD and LLVIP, **GRID** is initialized from the YOLOv8-World pretrained model.

Evaluation metrics. We employ the standard mean Average Precision (mAP) at an Intersection-over-Union (IoU) threshold of 0.5, consistent with the dataset’s official benchmarking protocol. We also highlight critical nuances in evaluating oriented bounding boxes (OBBs), particularly in the context of YOLOv8-OBB, which adopts ProbIoU to model geometric uncertainty and approximate rotated overlaps. Unlike parameterized rotated rectangles (5 degrees of freedom: center, width, height, and angle), the DroneVehicle dataset provides native quadrilateral annotations (8 degrees of freedom via four corner coordinates). To ensure accurate and rigorous evaluation, we compute IoU directly between the ground-truth quadrilaterals and predicted rotated boxes, which are converted into quadrilateral form. This approach mitigates the inaccuracies introduced by rotation parameterization and adheres to the evaluation standards of the DOTA benchmark [31], where rotated object detection is assessed using the official DOTA devkit. This setup ensures consistency with aerial image detection benchmarks and enables reliable and interpretable mAP₅₀ measurements for the OBB task.

4.2. Comparisons with State-of-the-Art Methods

Compared methods. We compare our method, *GRID*, with multi-modality (RGB+IR) detection methods. On the DroneVehicle dataset, we evaluate against YOLOv8 [22], UA-CMDet [10], TSFADet [32], CALNet [8], C²Former [33], CAGTDet [34], OAFA [35], E2E-MFD [36], DMM [9], M²D-LIF [37], DEGF-YOLO [38], and ADMPPF [17]. On the M3FD dataset, they are SDNet [39], TarDAL [26], CDDFuse [40], TINet [41], MMI-Det [42], ICAFusion [16], EME [43], FD²Net [44], MMFN [45], and EI²Det [46]. On the LLVIP dataset, CAMF [47], TFNet [48], MetaFusion [49], Diff-IF [50], ProbEn [12], CALNet [8], CSAA [51] TFDet [2], YOLO-Adaptor [52], and FD²Net [44] are compared. The results of these methods are obtained from their respective publications.

Quantitative comparison. Tables 1-3 report the performance of our proposed *GRID* framework compared with state-of-the-art (SOTA) methods on the DroneVehicle, M3FD, and LLVIP datasets. We evaluate three variants: *GRID-S*, *GRID-M*, and *GRID-L*. There are the following conclusions: (1) **Consistent performance gains across benchmarks.** *GRID-L* achieves the best overall performance on these datasets, particularly excelling in challenging categories such as *truck*, *van*, and *motorcycle*.

Table 1: Comparison of **GRID** with single-modal and multi-modal methods on the **DroneVehicle** test and val set. All the results are evaluated on **OBB** task. Best results are marked with **bold**.

Method	Basic Detector	DroneVehicle test						DroneVehicle val					
		Car	Truck	Fre	Bus	Van	mAP ₅₀	Car	Truck	Fre	Bus	Van	mAP ₅₀
YOLOv8s [22]	–	97.6	74.3	58.1	94.7	51.8	75.3	97.6	70.5	66.5	95.5	57.0	77.4
YOLOv8m [22]	–	97.9	77.3	61.6	95.0	53.4	77.0	97.8	74.4	67.9	95.4	59.7	79.1
YOLOv8l [22]	–	98.0	77.2	61.8	95.3	58.4	78.1	97.8	74.4	68.2	96.0	61.4	79.6
UA-CMDet [10]	RoITrans	87.5	60.7	46.8	87.1	38.0	64.0	87.5	60.7	46.8	87.1	38.0	64.0
TSFADet [32]	OR-CNN	89.2	72.0	54.2	88.1	48.8	70.4	89.9	67.9	63.7	89.8	54.0	73.1
CALNet [8]	YOLOv5l	90.3	76.2	63.0	89.1	58.5	75.4	90.3	73.7	68.7	89.7	59.7	76.4
C ² Former [33]	S ² ANet	90.0	72.1	57.6	88.7	55.4	72.8	90.2	68.3	64.4	89.8	58.5	74.2
CAGTDet [34]	OR-CNN	–	–	–	–	–	–	90.8	69.7	66.3	90.5	55.6	74.6
OAFa [35]	SLBAF	–	–	–	–	–	–	90.3	76.8	73.3	90.3	66.0	79.4
E2E-MFD [36]	YOLOv5s	90.3	79.3	64.6	89.8	63.1	77.4	–	–	–	–	–	–
DMM [9]	S ² ANet	90.4	79.8	68.2	89.9	68.6	79.4	–	–	–	–	–	–
M ² D-LIF [37]	YOLOv8m	97.8	81.0	67.9	96.0	64.6	81.4	98.1	81.6	76.5	96.4	69.7	84.5
DEGF-YOLO [38]	YOLOv8s	–	–	–	–	–	–	98.1	83.6	71.3	96.4	66.3	83.5
ADMPF [17]	YOLOv8s	97.9	82.3	69.3	95.9	64.8	82.1	–	–	–	–	–	–
GRID-S	YOLOv8s	98.2	82.5	69.3	96.6	66.0	82.5	98.3	82.1	78.7	96.6	68.7	84.9
GRID-M	YOLOv8m	98.3	83.2	70.6	96.7	66.3	83.0	98.4	83.6	78.6	96.9	70.3	85.5
GRID-L	YOLOv8l	98.3	83.8	69.9	96.7	67.9	83.3	98.3	83.2	78.3	97.2	70.9	85.6

This highlights the effectiveness of our method in enhancing cross-modal feature representation under difficult conditions. It should be noted that several prior works on the M3FD dataset do not explicitly report the backbone detectors used, making it difficult to present a strictly detector-aligned comparison. Despite the overall performance gains, **GRID-L** underperforms in the *lamp* category (84.7%) compared to FD²Net (87.8%), which may be attributed to insufficient textual priors for this category as well as the interference of IR thermal noise that disrupts inter-modal spatial alignment. Additionally, unlike DroneVehicle and M3FD, the LLVIP dataset has only one category (person) and a few small objects, which limits the use of analogical reasoning and re-

Table 2: Comprehensive experiments on **M3FD** dataset.

Method	Peo	Bus	Car	Mot	Lam	Tru	mAP ₅₀	mAP
SDNet [39]	79.5	81.0	92.6	67.1	84.2	69.4	79.0	–
TarDAL [26]	78.0	82.9	87.7	68.6	76.9	81.8	80.7	50.1
CDDFuse [40]	81.1	82.3	93.2	74.0	87.7	72.7	81.9	–
TINet [41]	78.2	83.1	87.5	68.3	78.5	81.6	79.8	49.5
MMI-Det [42]	80.6	89.5	90.7	70.7	83.3	85.9	83.5	51.9
ICAFusion [16]	83.0	92.3	91.0	76.5	85.0	88.9	85.1	53.5
EME [43]	79.9	89.2	91.8	76.2	78.3	85.3	83.4	–
FD ² Net [44]	83.7	82.7	93.6	78.1	87.8	73.8	83.5	–
MMFN [45]	83.0	92.1	93.2	73.7	87.6	87.4	86.2	–
El ² Det [46]	82.8	91.6	91.4	77.2	84.7	89.3	86.2	55.5
GRID-S	86.2	91.3	93.0	77.9	82.1	87.3	86.3	63.1
GRID-M	87.4	92.8	93.5	80.6	82.8	88.9	87.7	65.3
GRID-L	88.2	94.5	94.1	79.9	84.7	91.0	88.7	67.0

sults in minimal performance gains. (2) **Favorable trade-off between accuracy and efficiency.** As shown in Table 4, on the DroneVehicle test dataset, **GRID-M** improves mAP₅₀ by +1.6% compared to M²D-LIF, despite introducing 12.3M more parameters. Meanwhile, **GRID-S** achieves a +0.4% gain over ADMPF with 7.4M fewer parameters and 19.8 higher FPS, demonstrating strong efficiency without sacrificing accuracy. Overall, the **GRID** framework offers a compelling balance between accuracy, model complexity, and real-time performance, making it a strong candidate for deployment in practical RGB-IR object detection scenarios.

4.3. Further studies

Ablation study. To evaluate the effectiveness of the key components in our **GRID** framework, we perform an ablation study on the DroneVehicle test set. The baseline (B) is a single-stage dual-stream detector [29] employing three fusion operations with simple addition. We incrementally integrate the Language Guided Objectness Attention (LGOA) module, which enhances semantic localization and mitigates pixel-

Table 3: Comprehensive experiments on **LLVIP** dataset.

Method	Basic Detector	mAP ₅₀	mAP
CAMF [47]	YOLOv5	89.0	55.6
TFNet [48]	Faster-RCNN	–	57.6
MetaFusion [49]	YOLOv5	91.0	56.9
Diff-IF [50]	YOLOv8	93.3	59.5
ProbEn [12]	Faster-RCNN	93.4	51.5
CALNet [8]	YOLOv5l	–	63.4
CSAA [51]	Faster-RCNN	94.3	59.2
TFDet [2]	Faster-RCNN	96.0	59.4
YOLO-Adaptor [52]	YOLOv5	96.5	–
FD ² Net [44]	YOLOv5	96.2	70.0
GRID-S	YOLOv8s	96.1	70.1
GRID-M	YOLOv8m	96.7	71.5
GRID-L	YOLOv5l	96.6	71.3
GRID-L	YOLOv8l	96.9	72.0

level misalignment; Intra-Modal Analogical Reasoning (Intra-AR) block, which re-
 350 fines modality-specific features using semantic relations; and Inter-Modal Spatial Ana-
 logical Reasoning (Inter-AR), which captures cross-modal spatial relations to transfer
 cross-position information. Intra-AR and Inter-AR together constitute the two stages
 of analogical reasoning.

As shown in Table 5, each module contributes positively to detection performance.
 355 Incorporating LGOA yields a 2.2% gain in mAP₅₀ over the baseline (from 77.0% to
 79.2%), demonstrating its efficacy in leveraging language semantics for precise object
 localization. Adding Intra-AR further improves performance to 82.0%, underscoring
 its role in enhancing modality-specific feature refinement. Substituting Intra-AR with
 Inter-AR results in a slightly higher mAP₅₀ of 82.3%, highlighting the importance
 360 of cross-modal spatial context. Combining both Intra-AR and Inter-AR in the full
GRID framework achieves the best performance (83.0%), surpassing the Intra-AR-

Table 4: Comparison of computational efficiency and performance metrics for different detection algorithms.

Method	Basic Detector	Param↓	FPS↑	mAP ₅₀
UA-CMDet [10]	RoITrans	139.7M	9.1	64.0
TSFADet [32]	OR-CNN	104.7M	–	70.4
CALNet [8]	YOLOv5l	313.0M	–	75.4
C ² Former [33]	S ² ANet	132.5M	12.9	72.8
DMM [9]	S ² ANet	87.97M	–	79.4
M ² D-LIF [37]	YOLOv8m	37.1M	–	81.4
ADMPF [17]	YOLOv8s	31.8M	39.8	82.1
GRID-S	YOLOv8s	24.4M	59.6	82.5
GRID-M	YOLOv8m	49.4M	42.8	83.0
GRID-L	YOLOv8l	77.2M	31.9	83.3

Table 5: Ablation study on DroneVehicle test set.

Methods	LGOA	Intra-AR	Inter-AR	mAP ₅₀
Baseline (B)	×	×	×	77.0
B+LGOA	√	×	×	79.2
B+LGOA+Intra-AR	√	√	×	82.0
B+LGOA+Inter-AR	√	×	√	82.3
GRID-M	√	√	√	83.0

only and Inter-AR-only variants by 1.0% and 0.7%, respectively. To further validate these contributions, we visualize feature maps in Fig. 3, where **GRID** exhibits more distinct activation regions. These results validate the synergistic effect of two-stage
 365 analogical reasoning in capturing semantic and spatial cues for robust multi-modal object detection.

Evidence of analogical reasoning works. To validate the rationality of the proposed semantic-spatial analogical reasoning, we conduct ablation studies and visual analyses to investigate its impact on multi-scale object detection. As shown in Table 6,
 370 the proposed **GRID** module substantially improves performance, particularly on small

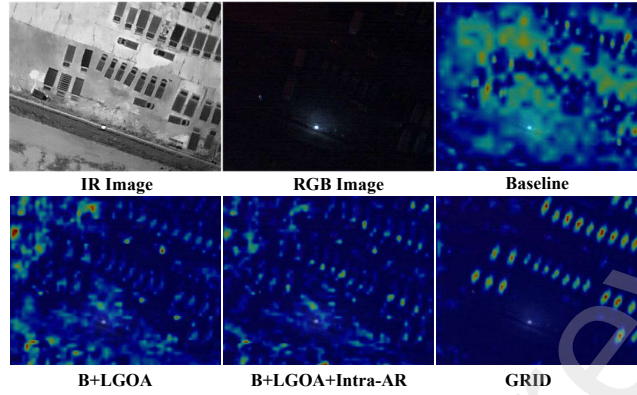


Figure 3: Visualization of feature maps in Baseline (B), B+LGOA, B+LGOA+Intra-AR, and GRID.

Table 6: Evidence of analogical reasoning on M3FD dataset.

Method	mAPs	mAPm	mAPI
Baseline	21.0	60.7	74.2
GRID	35.3	68.4	81.1

objects (mAPs: 21.0% $\rightarrow$ 35.3%), indicating its strong capability to enhance detection under challenging conditions. This improvement supports our hypothesis that semantically similar large objects can serve as effective analogical evidence for refining small object features. Fig. 4 further illustrates this effect. In dense scenes, **GRID** successfully detects small objects that are missed by the baseline model. This is attributed to the contextual and visual similarity shared among objects of the same category, where large and clearly visible instances provide valuable cues for identifying smaller or occluded ones. The attention heatmaps show that the baseline model tends to focus on dispersed regions, whereas **GRID** effectively attends to the small targets, highlighting the semantic transfer process. This demonstrates how the model leverages informative regions across modalities and scales. Overall, these results confirm that our analogical reasoning mechanism not only enhances intra-modal contextual understanding but also facilitates cross-modal spatial transfer, thereby improving both detection robustness and granularity.

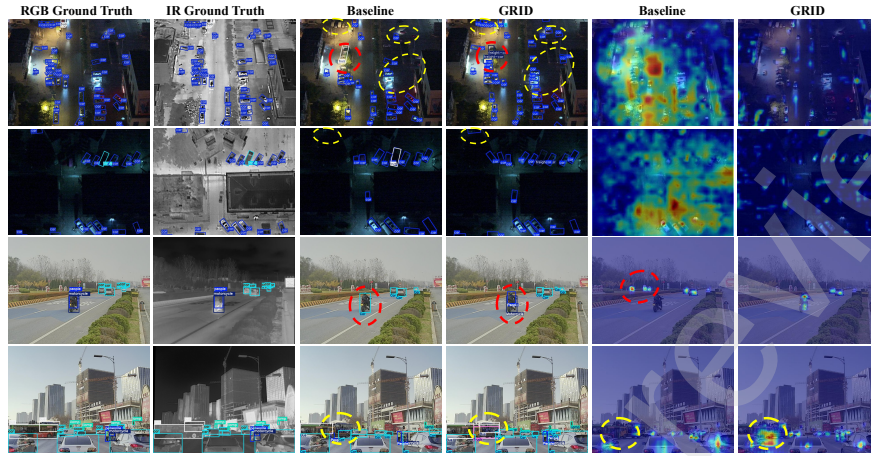


Figure 4: Visual evidence supporting analogical reasoning in **GRID** compared to baseline. Yellow dashed boxes highlight objects missed by the baseline but correctly detected by **GRID**. Red dashed boxes indicate false positives from the baseline that are correctly identified or ignored by **GRID**. Columns 3 and 4 show detection results, while columns 5 and 6 present visualizations of the corresponding feature maps. *Zoom in for best view.*

385 **Two-stage analogical reasoning vs. full graph reasoning.** To validate the effectiveness of the two-stage S2AR framework, we compare it against a simplified variant, Full Graph Reasoning (FGR), which performs analogical reasoning on a full graph. Fig. 5 illustrates S2AR’s superior performance across the DroneVehicle and M3FD datasets, whether from an overall average or individual per-class performance. These
390 gains are attributed to S2AR’s two-stage reasoning design, which mitigates noise from heterogeneous features. In contrast, full graph convolution suffers from modality interference, reducing feature discriminability, particularly for challenging categories like *van* and *truck*. These results highlight the advantage of a staged fusion strategy.

395 **Analysis of placement positions.** To investigate the impact of the placement position of the **GRID** module in the backbone, we insert it after feature extraction stages 1, 2, and 3 individually, as well as across all stages. Table 7 reports that placing **GRID** at any single stage enhances mAP₅₀, achieving 82.6% (stage-1), 82.1% (stage-2), and 82.4% (stage-3), all matching or surpassing the SOTA ADMPPF method (82.1%) [17]. The optimal configuration, applying **GRID** across all stages, attains an mAP₅₀ of

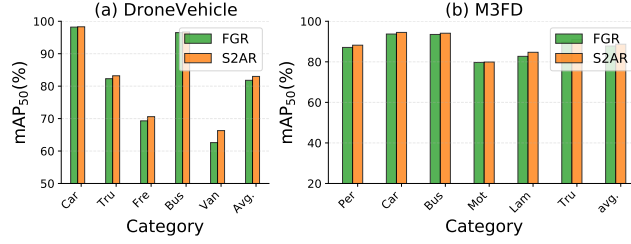


Figure 5: Performance comparison of two-stage analogical reasoning and Full Graph Reasoning (FGR).

Table 7: Analysis of placing GRID module at different feature extraction stages on the DroneVehicle test set.

Stage	Car	Truck	Fre	Bus	Van	mAP ₅₀
stage-1	98.2	83	69.9	96.5	65.5	82.6
stage-2	98.2	81.7	68.9	96.5	65.4	82.1
stage-3	98.2	82.3	69.4	96.3	65.6	82.4
All	98.3	83.2	70.6	96.7	66.3	83.0

83.0%. This superior performance stems from the S2AR framework’s ability to integrate multi-scale features, combining fine-grained details from early stages with semantically rich representations from later stages. Stage-1 placement yields the second-best mAP₅₀, benefiting from higher-resolution features and more graph nodes, which enable richer contextual reasoning. These results underscore the advantage of multi-stage *GRID* integration for robust multispectral object detection.

Analysis of cross-modal progressive text enhancement and Prompt Design. To evaluate the effectiveness of the proposed progressive cross-modal text enhancement strategy, we conduct a comparative study between two settings: with and without progressive text enhancement. In the w/o progressive enhancement variant, each layer directly utilizes static text embeddings generated by the CLIP text encoder, without incorporating semantic refinements from preceding layers. As reported in Table 8, removing the progressive enhancement mechanism leads to consistent performance degradation across all categories, with an overall drop in mAP₅₀. This decline can be attributed to the inherent appearance discrepancies between RGB and IR modalities,

Table 8: Comparison of w/o Progressive Text Enhancement (PTE), Learnable Prompts (LP) and GRID on DroneVehicle dataset.

Method	Car	Truck	Fre	Bus	Van	mAP ₅₀
w/o PTE	98.2	83.2	69.7	96.5	64.5	82.4
LP	98.3	83.0	70.4	96.1	66.8	82.9
GRID	98.3	83.2	70.6	96.7	66.3	83.0

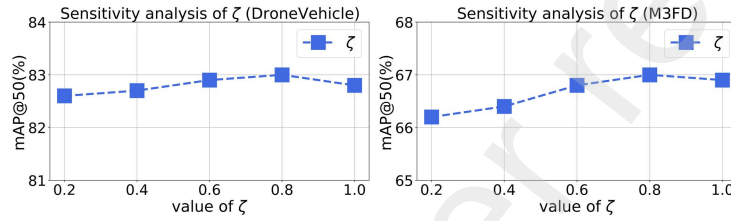


Figure 6: Sensitivity analysis of ζ .

415 which are insufficiently addressed by static textual representations. In contrast, the proposed progressive enhancement strategy incrementally refines textual embeddings across network layers, facilitating better semantic alignment with multimodal visual features. This process promotes semantic consistency and enhances feature discriminability, thereby enabling the model to learn more robust and discriminative representations for RGB-IR object detection.

420 Furthermore, we investigate the impact of prompt design by comparing fixed prompts with learnable prompts. Specifically, **GRID** adopts a fixed prompt formulation, while LP employs learnable prompts that are optimized during training. As shown in Table 8, the performance gap between the two strategies is marginal, with **GRID** achieving a slightly higher mAP₅₀ (+0.1%) compared to LP. This observation suggests that the choice of prompt design has a limited influence on the overall detection performance.

430 **Sensitivity analysis.** We conduct a sensitivity analysis on the hyperparameter ζ , which balances the fusion between image and text features in the progressive cross-modal text enhancement block. As shown in Fig. 6, the detection performance (mAP₅₀) remains stable across a broad range of ζ values from 0.2 to 0.8, with the best perfor-

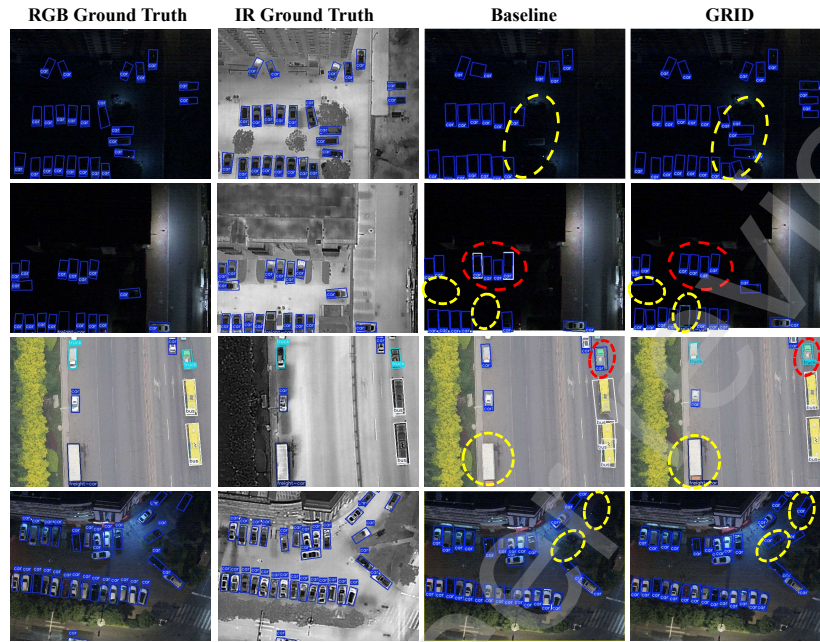


Figure 7: Visualization comparison on DroneVehicle. Our GRID is compared with Ground Truth and Baseline. Regions highlighted with yellow dashed boxes indicate instances that are successfully detected by GRID but missed by the baseline model. Red dashed boxes denote false positives by the baseline that are correctly suppressed by GRID. *Zoom in for best view.*

mance achieved at $\zeta = 0.8$. However, when ζ is increased to 1, performance begins to degrade, suggesting that an over-reliance on text features may hinder effective multi-modal fusion. These results indicate that appropriately tuning ζ is critical to achieving optimal detection performance across different datasets.

435 **Visualization.** The qualitative results are presented in Fig. 7–9. From left to right, each example shows the RGB ground truth, IR ground truth, detection results of the Baseline model, and our proposed **GRID** method. As highlighted by the yellow and
440 red dashed circles, **GRID** significantly enhances detection performance in complex scenes by increasing true positive (TP) detections and improving the confidence and accuracy of small or visually ambiguous object recognition. These results demonstrate **GRID**’s ability to leverage analogical reasoning to refine object representations across modalities and scales.

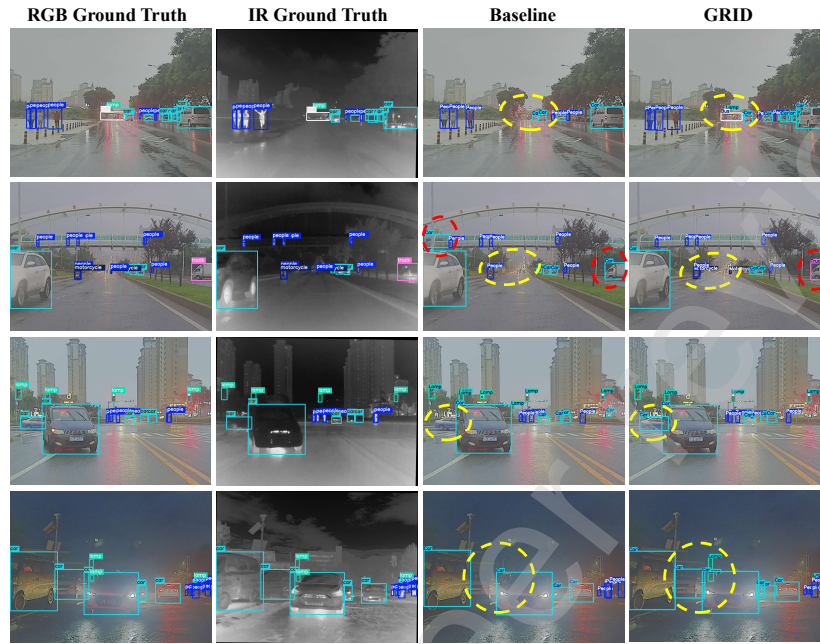


Figure 8: Visualization comparasion on M3FD. *Zoom in for best view.*

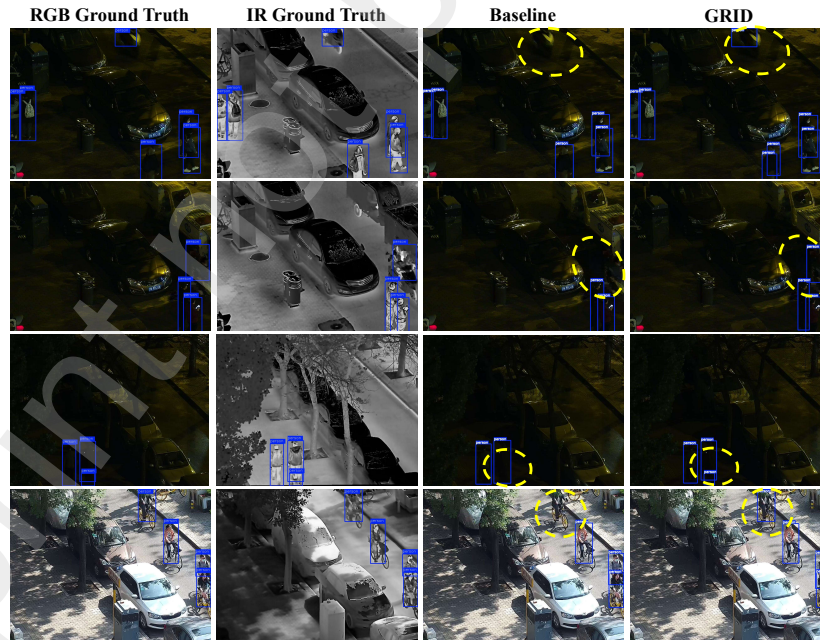


Figure 9: Visualization comparison on LLVIP. *Zoom in for best view.*

5. Conclusion

We proposed a novel framework for RGB-IR object detection, termed **GRID**, moving beyond traditional pixel-wise alignment approaches. **GRID** comprises two core modules: Language-Guided Objectness Attention (LGOA) and Semantic-Spatial Analogical Reasoning (S2AR). LGOA utilizes CLIP’s semantic priors to progressively enhance textual representations, which subsequently guide the generation of modality-specific objectness attention maps, thereby narrowing the gap between RGB and IR features. S2AR constructs a graph using text embeddings and objectness-filtered RGB/IR regions, models both semantic and spatial relations, and performs two-stage graph reasoning to refine feature representations. In this way, **GRID** achieves robust detection performance under challenging conditions such as low illumination and complex aerial scenes.

6. CRediT authorship contribution statement

Ting Li: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualization and Writing-original draft. **Maoyuan Zhao**: Data curation, Validation and Visualization. **Haiwen Xu**: Formal analysis, Writing-review & editing. **Hongkai Liu**: Software, Validation and Visualization. **Nianxin Li**: Methodology, and Validation. **Shuaifeng Li**: Methodology, Validation. **Mao Ye**: Funding acquisition, Project administration, Writing-review & editing.

7. Generative AI disclosure statement

The authors confirm that no generative artificial intelligence (AI) tools were used in the preparation of this manuscript.

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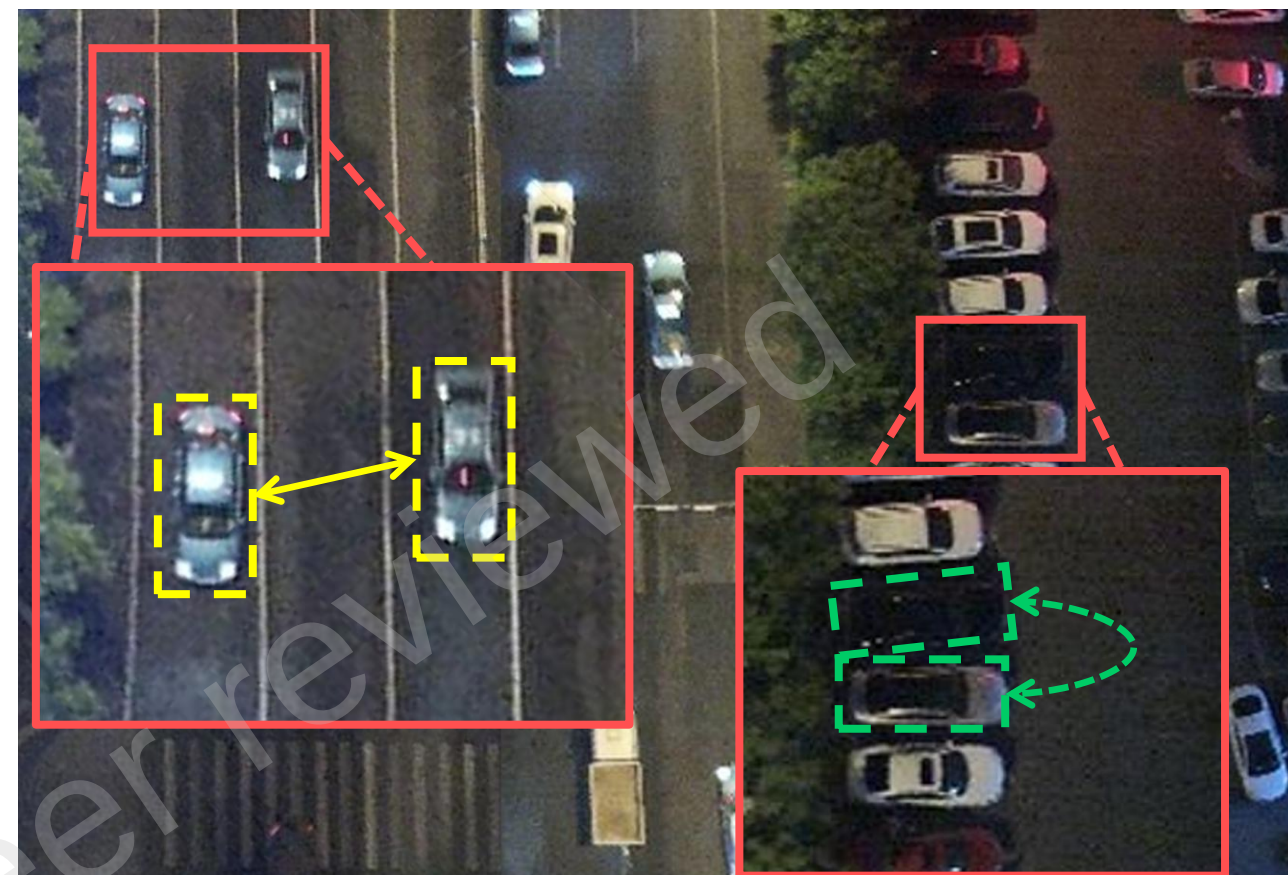
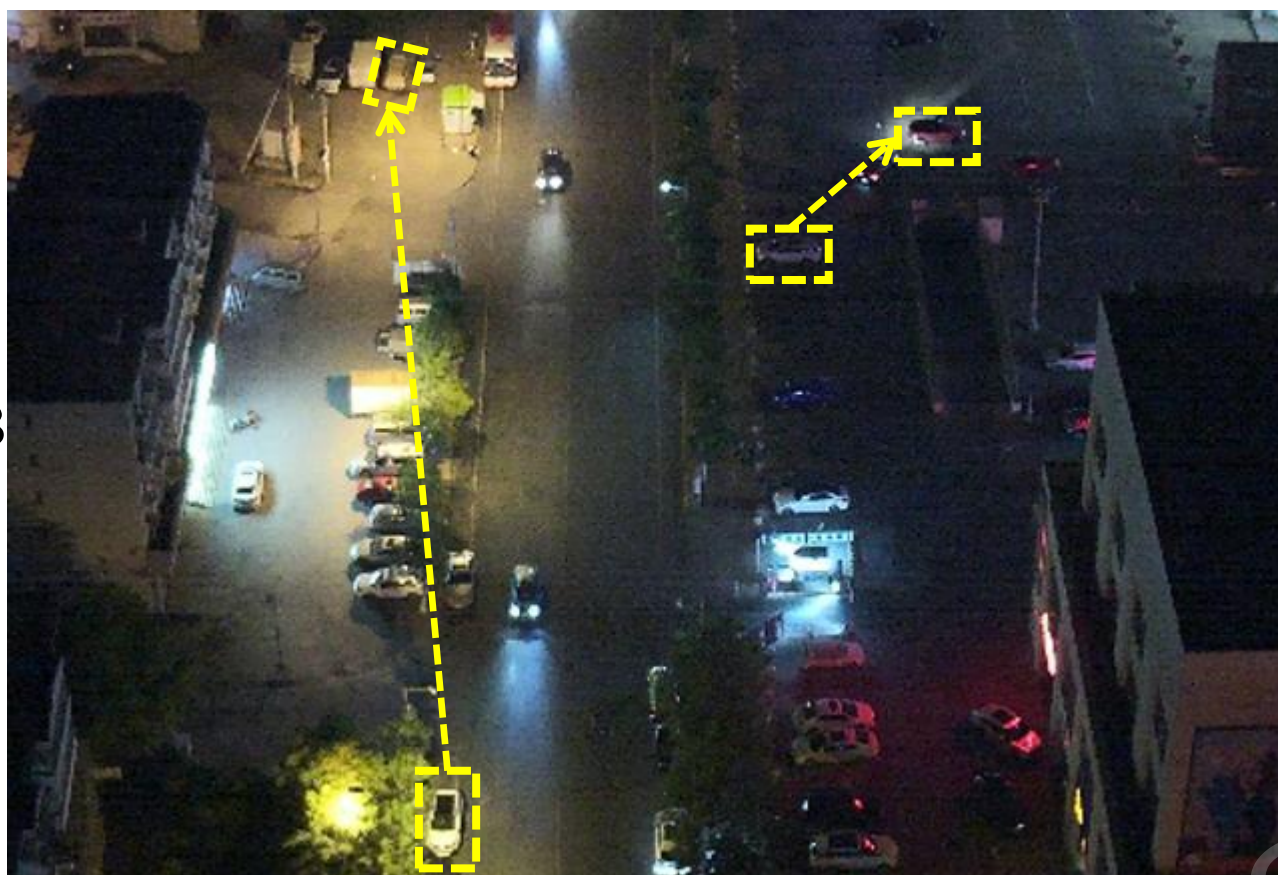
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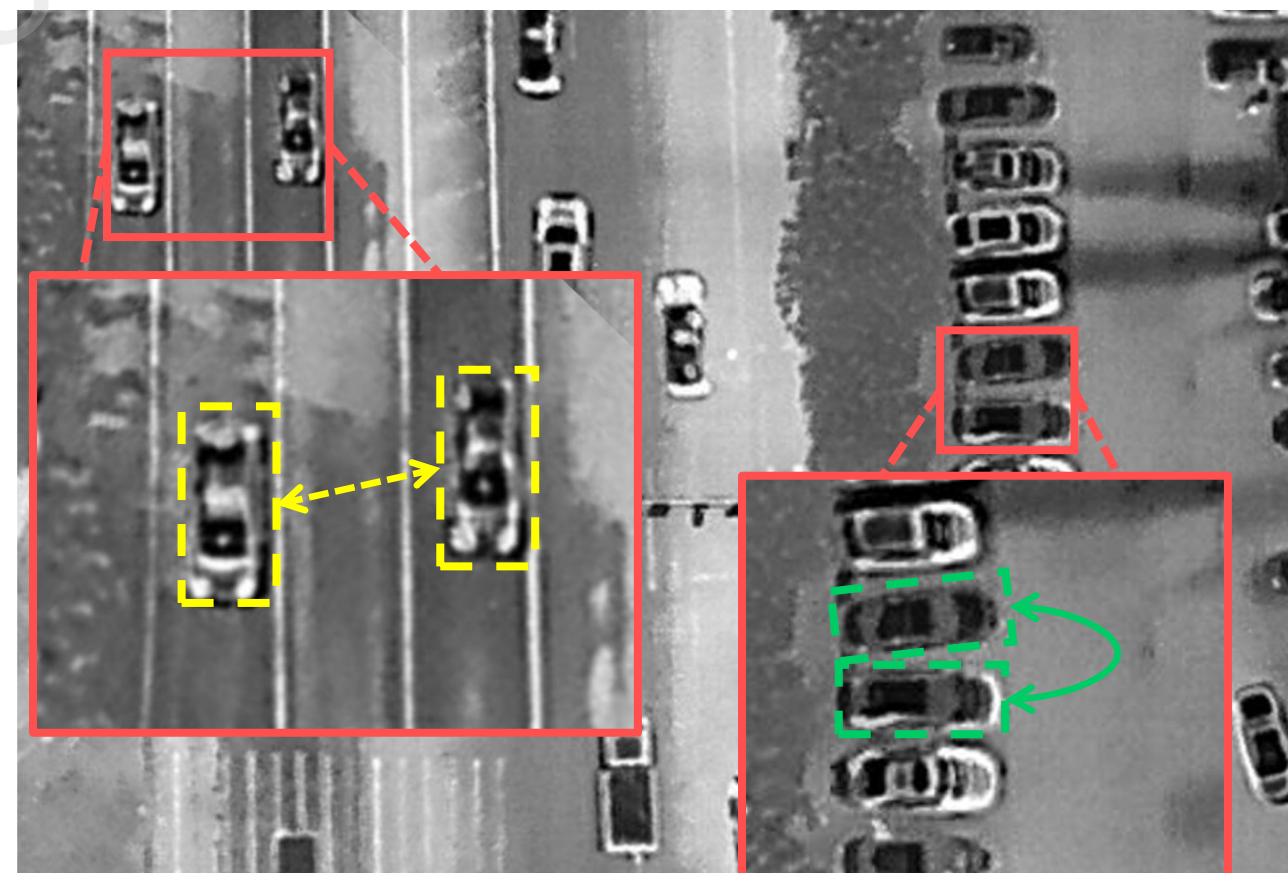
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RGB

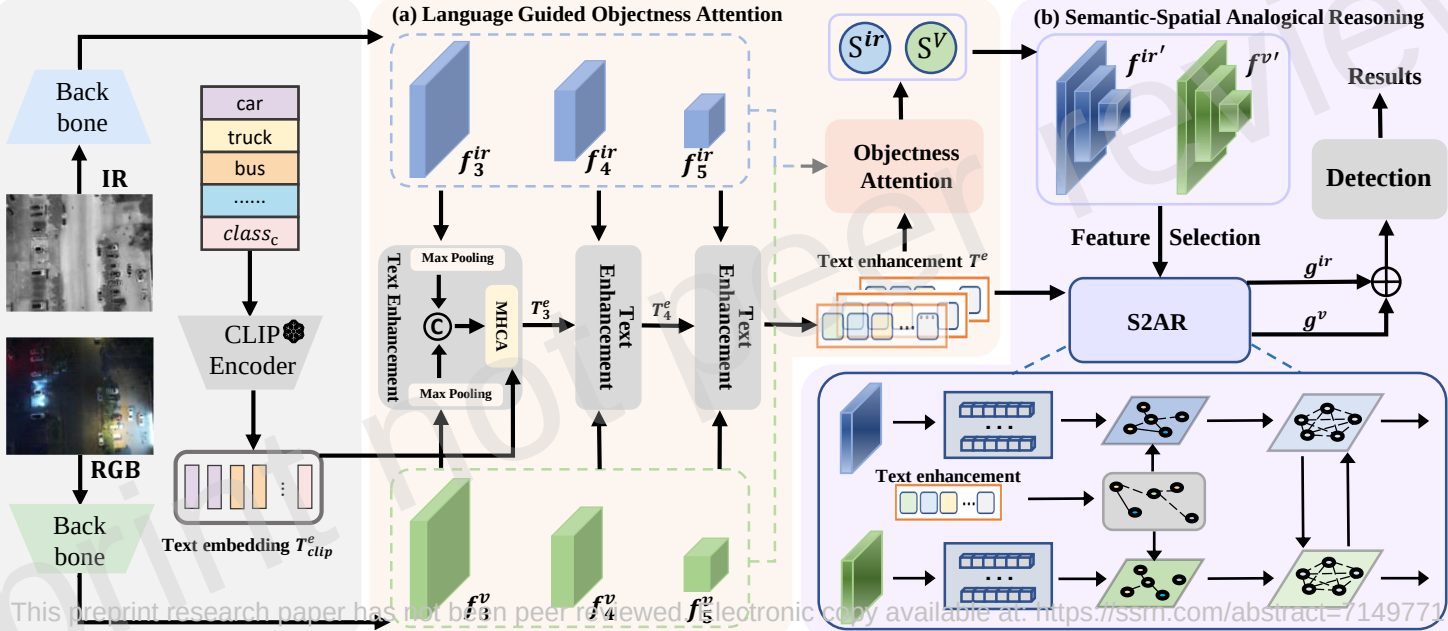


IR



(a) Semantic relation mapping

(b) Spatial relation mapping

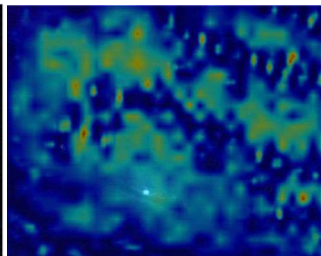




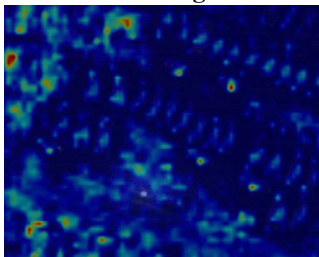
IR Image



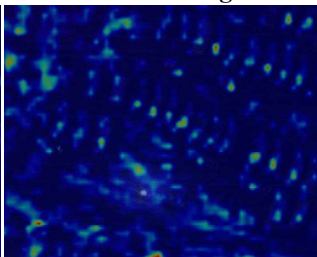
RGB Image



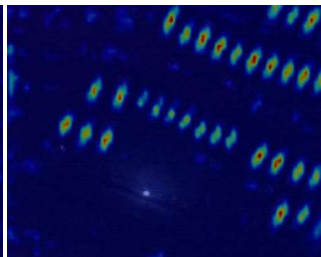
Baseline



B+LGOA

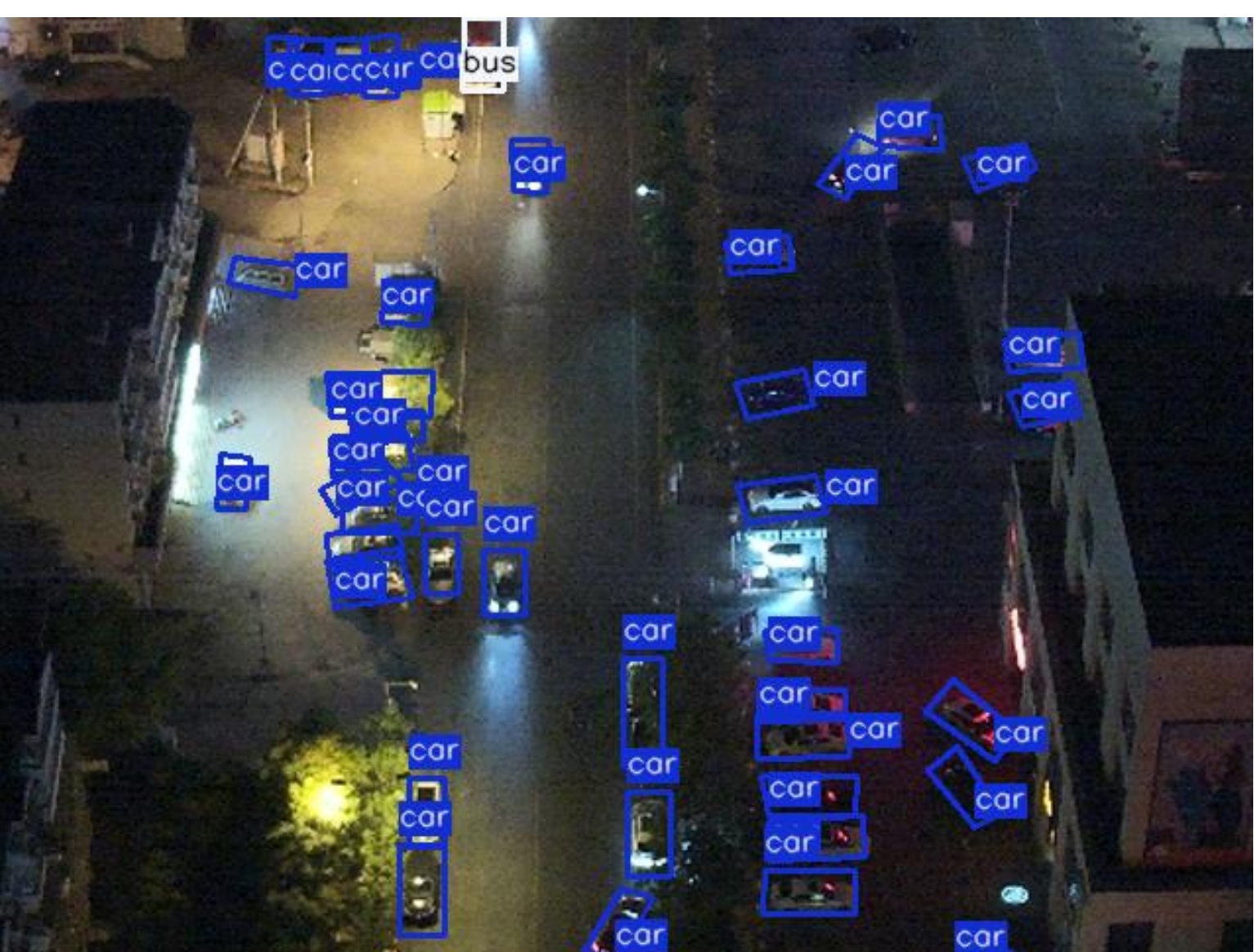


B+LGOA+Intra-AR



GRID

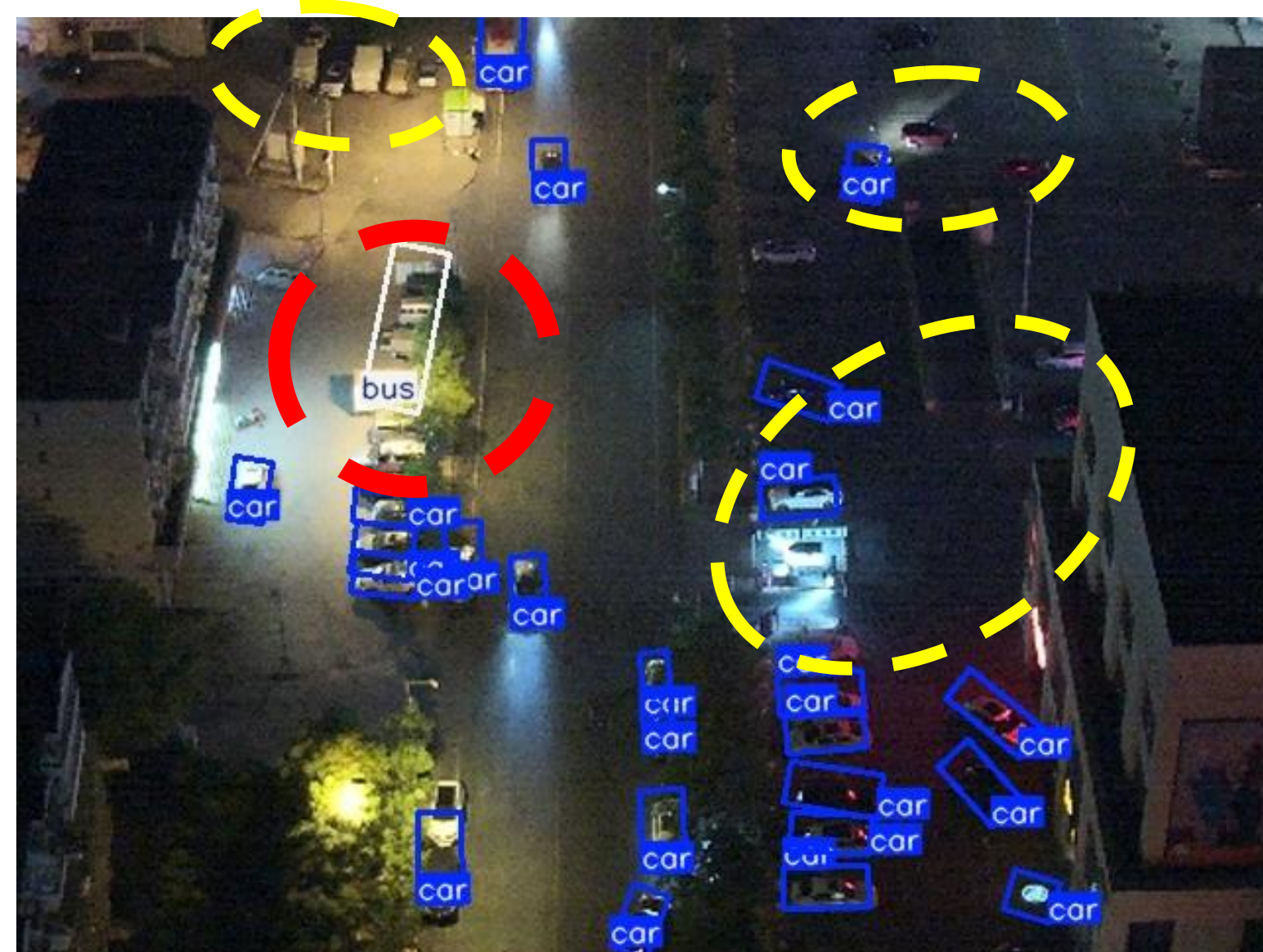
RGB Ground Truth



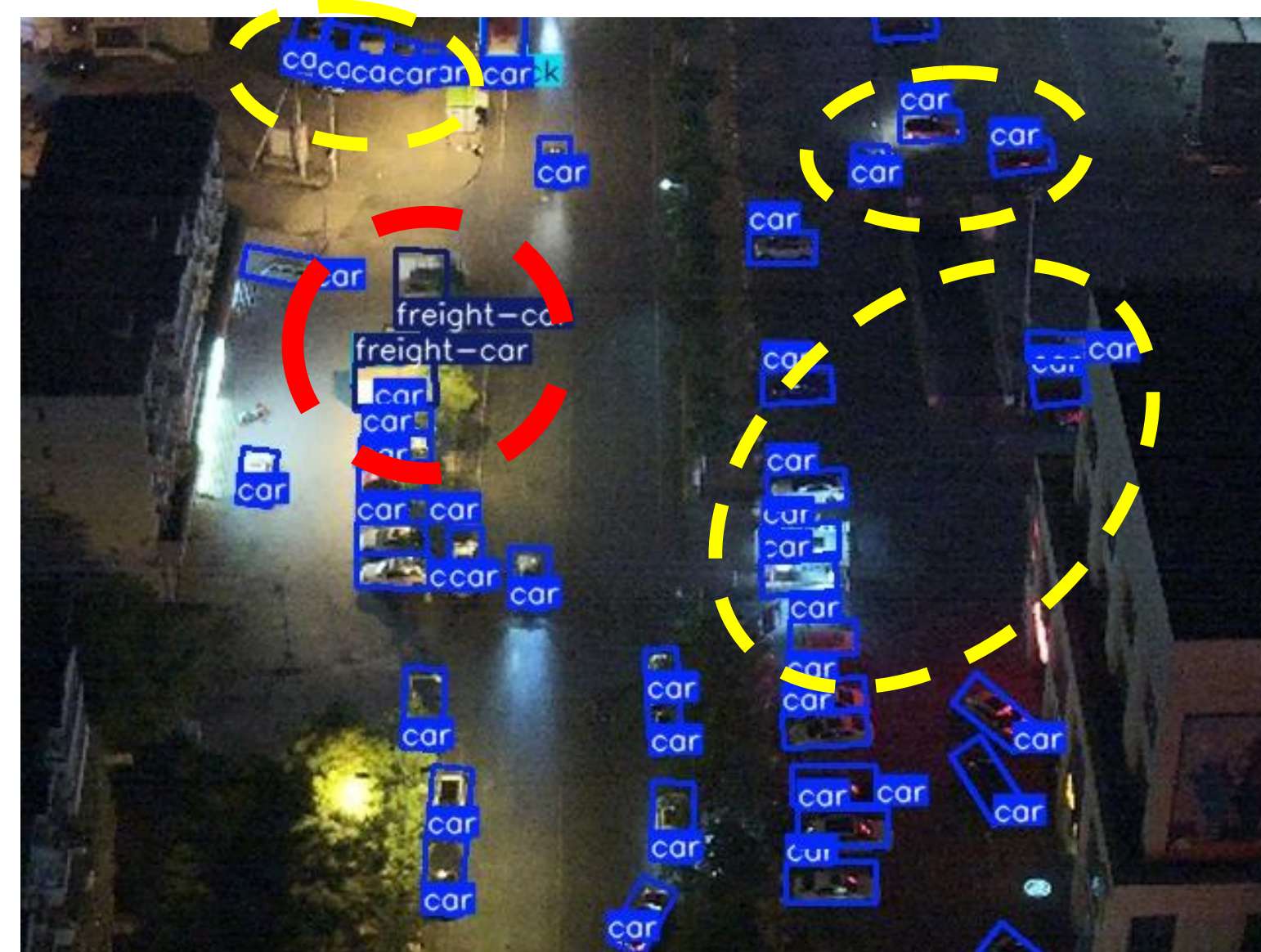
IR Ground Truth



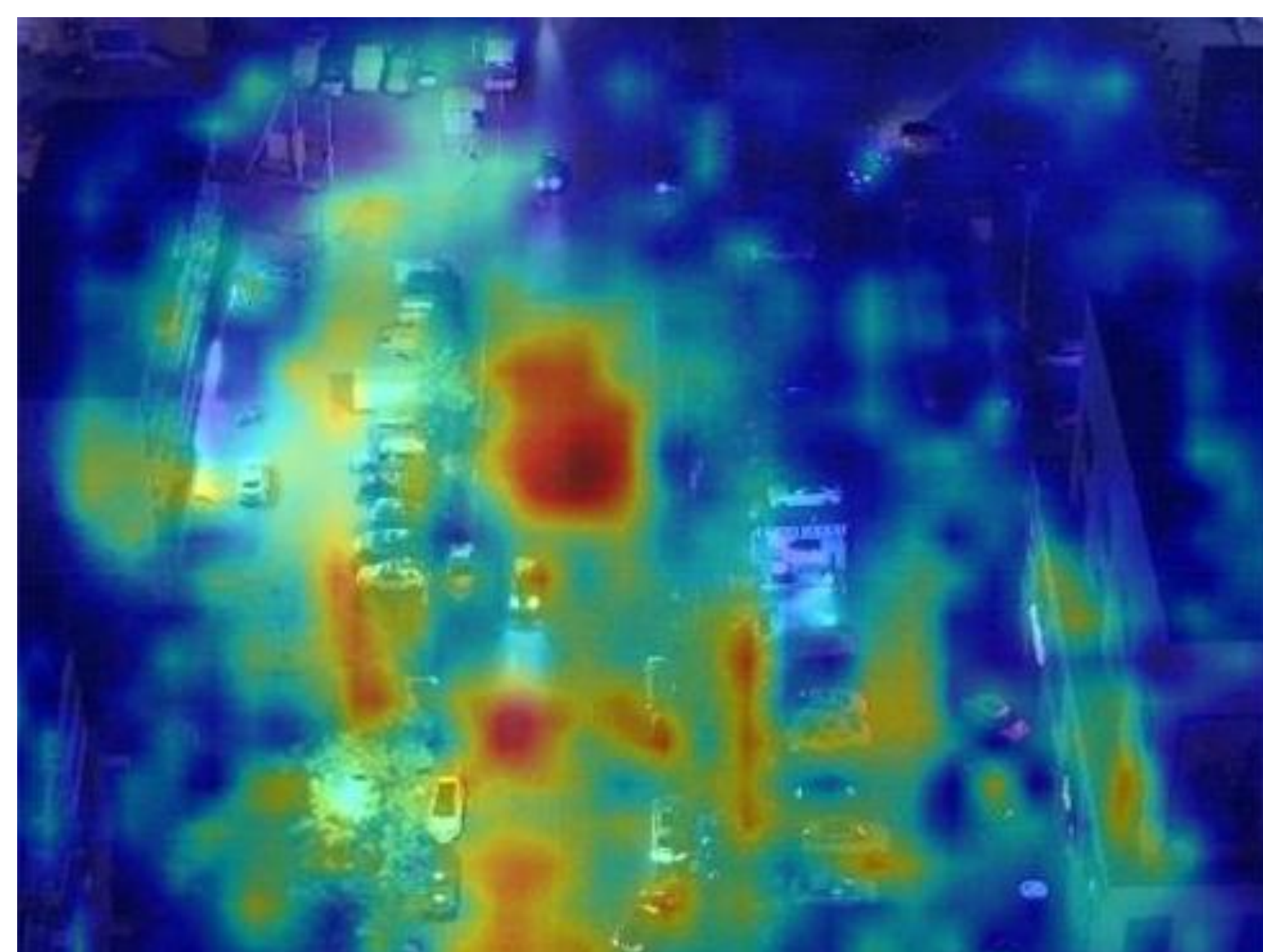
Baseline



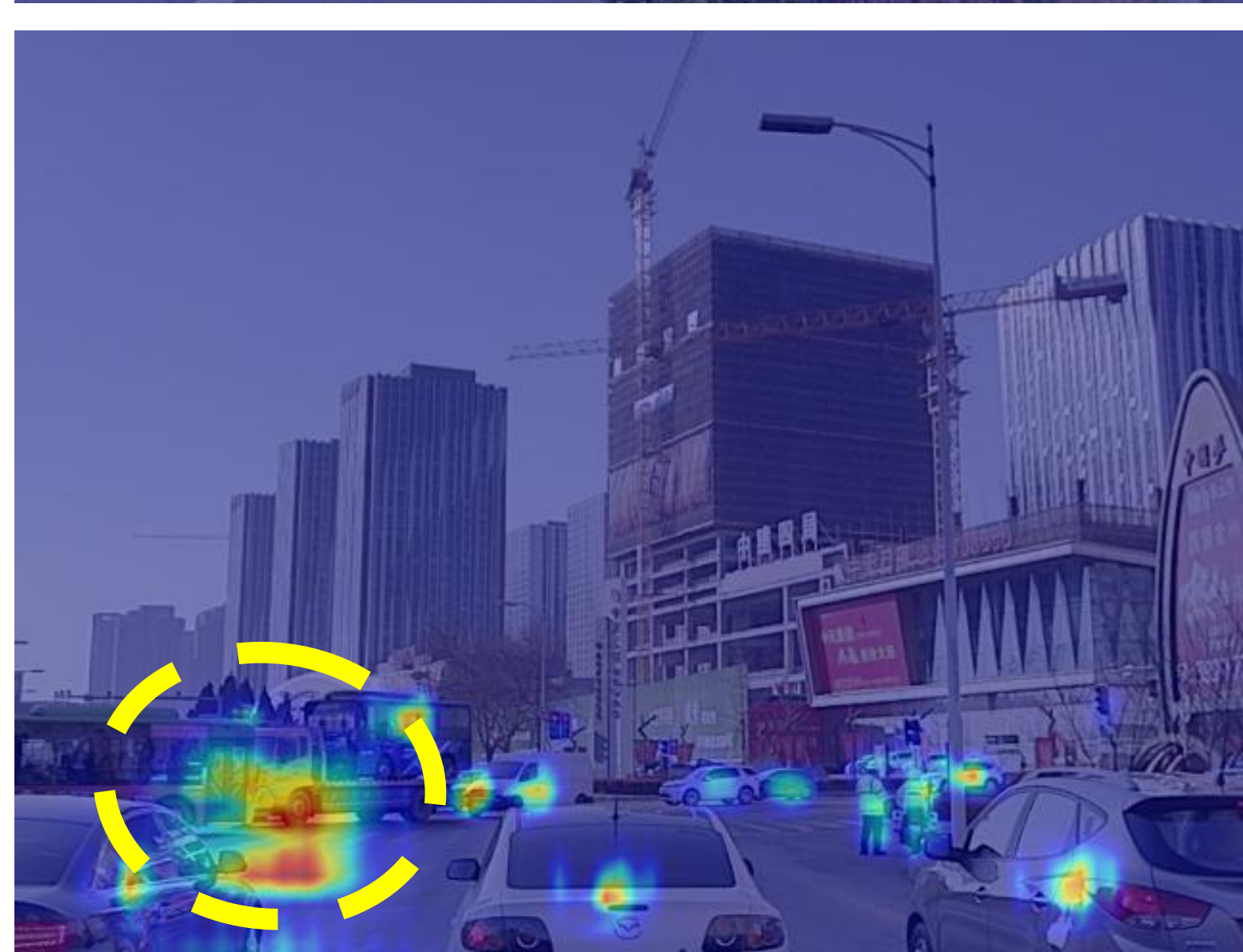
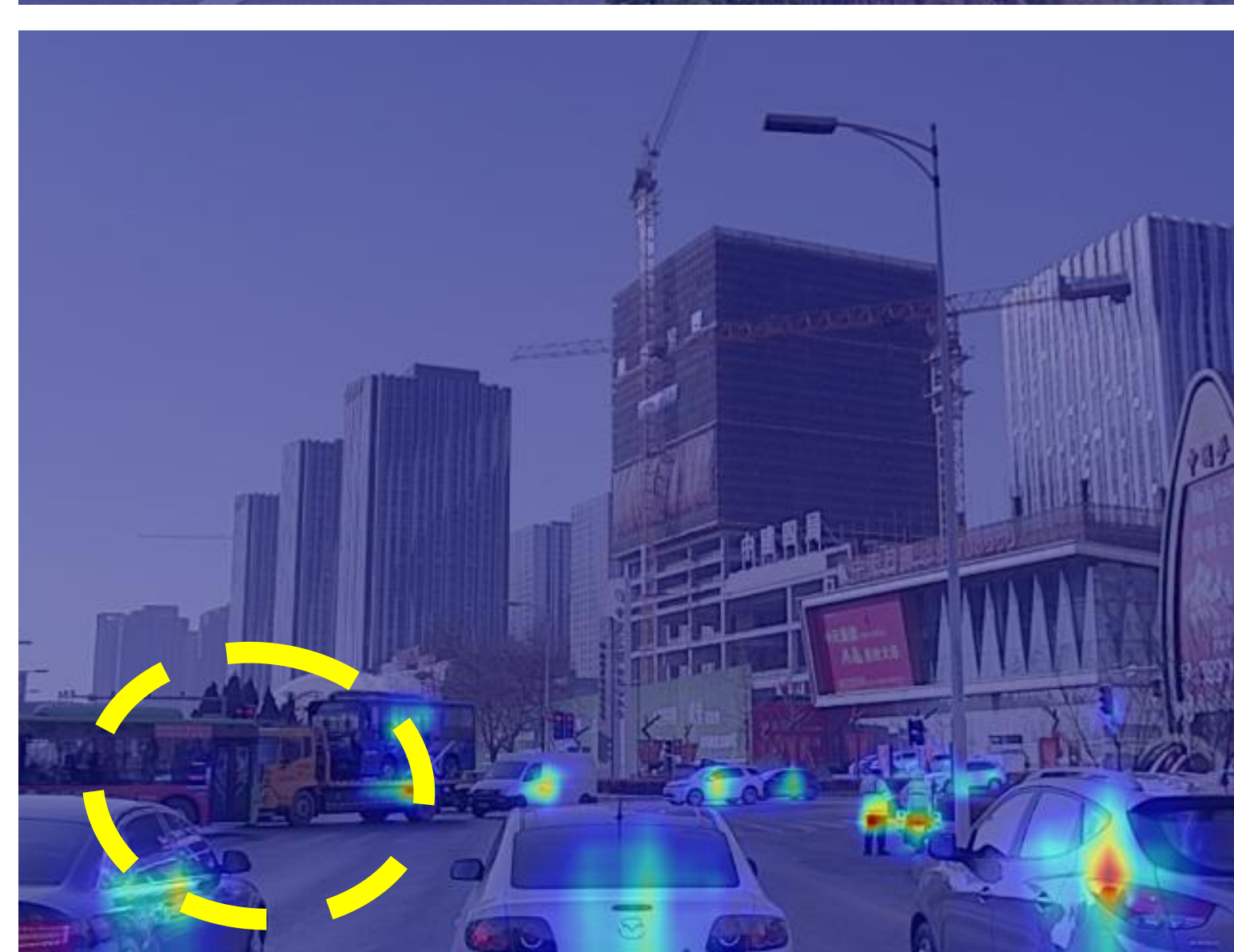
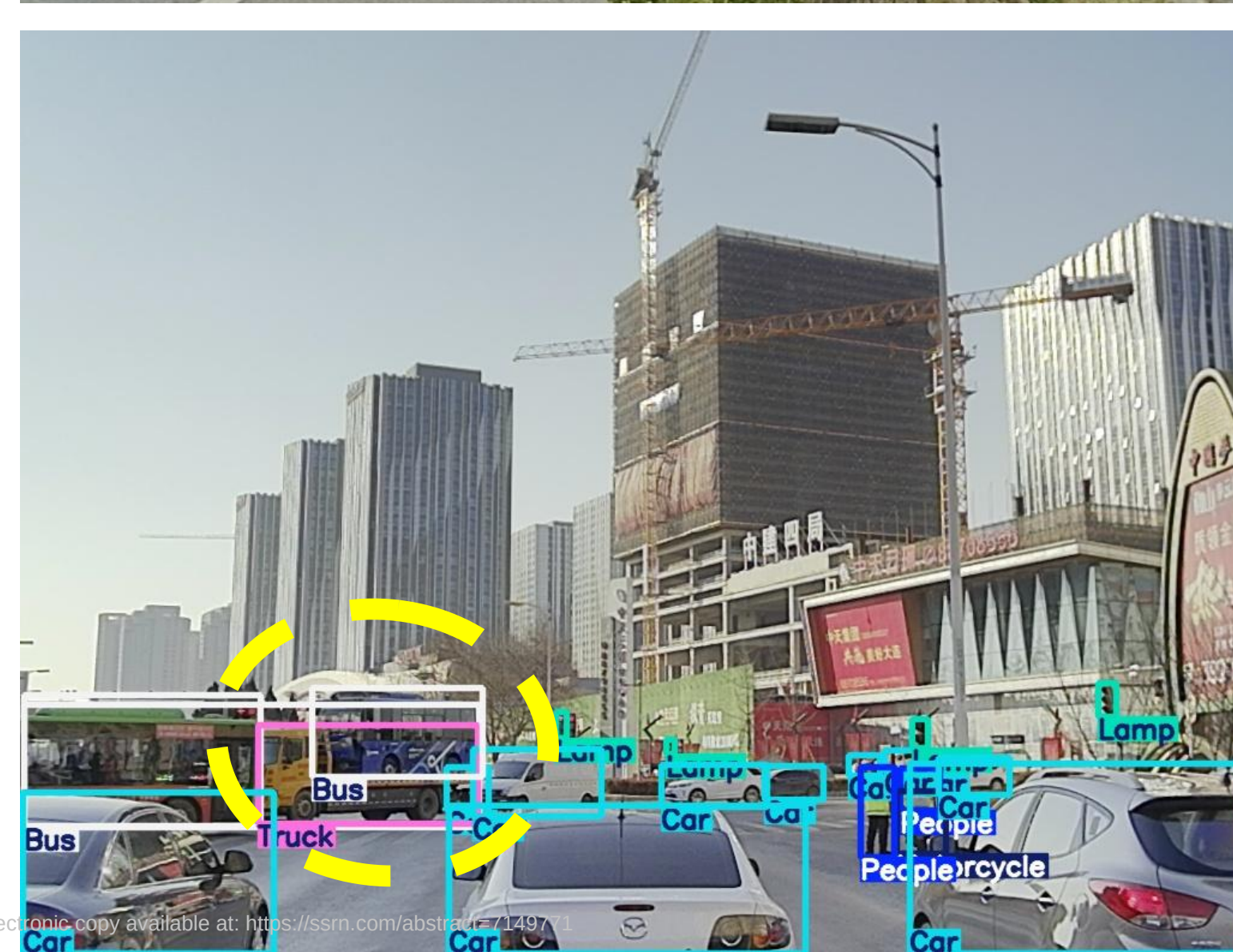
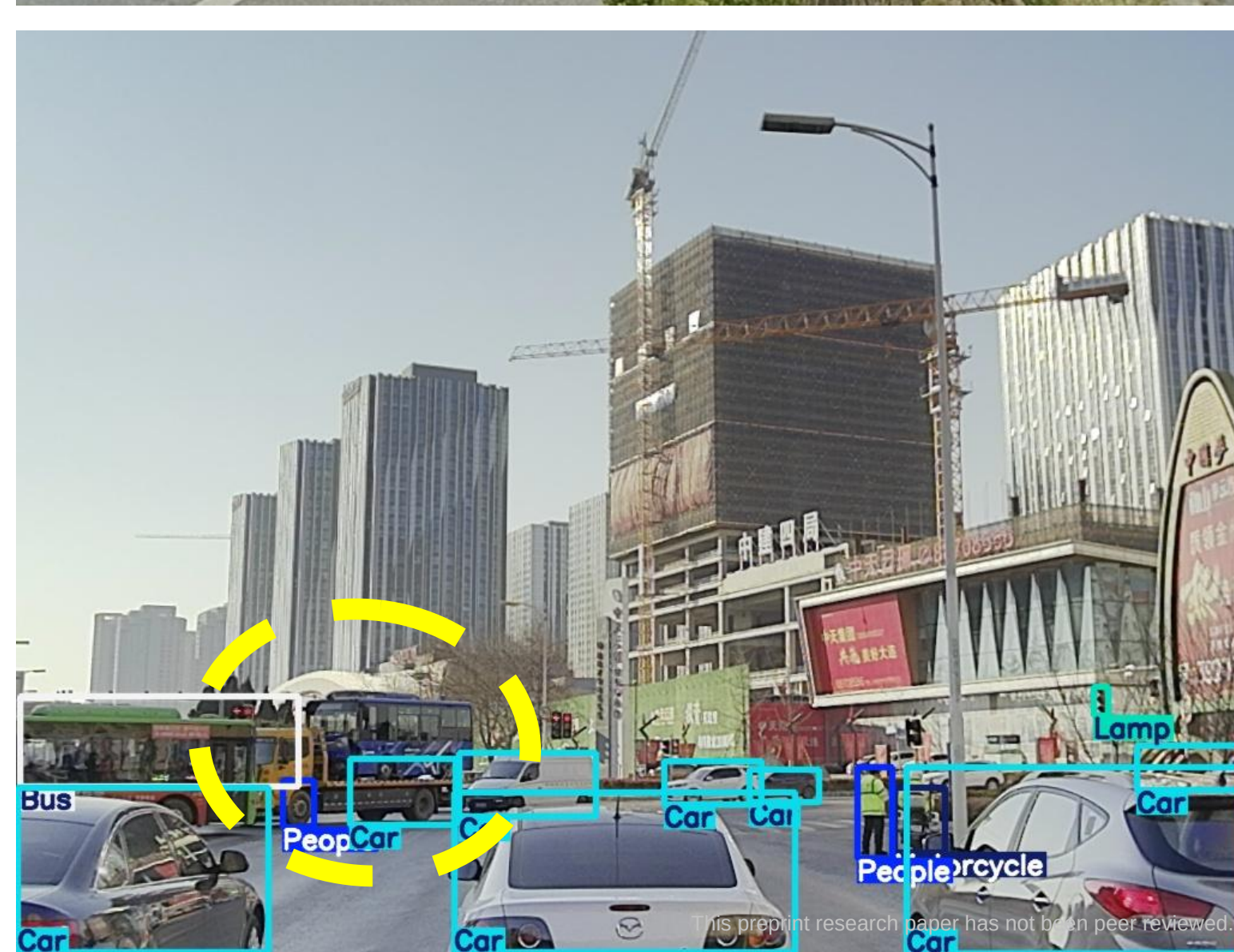
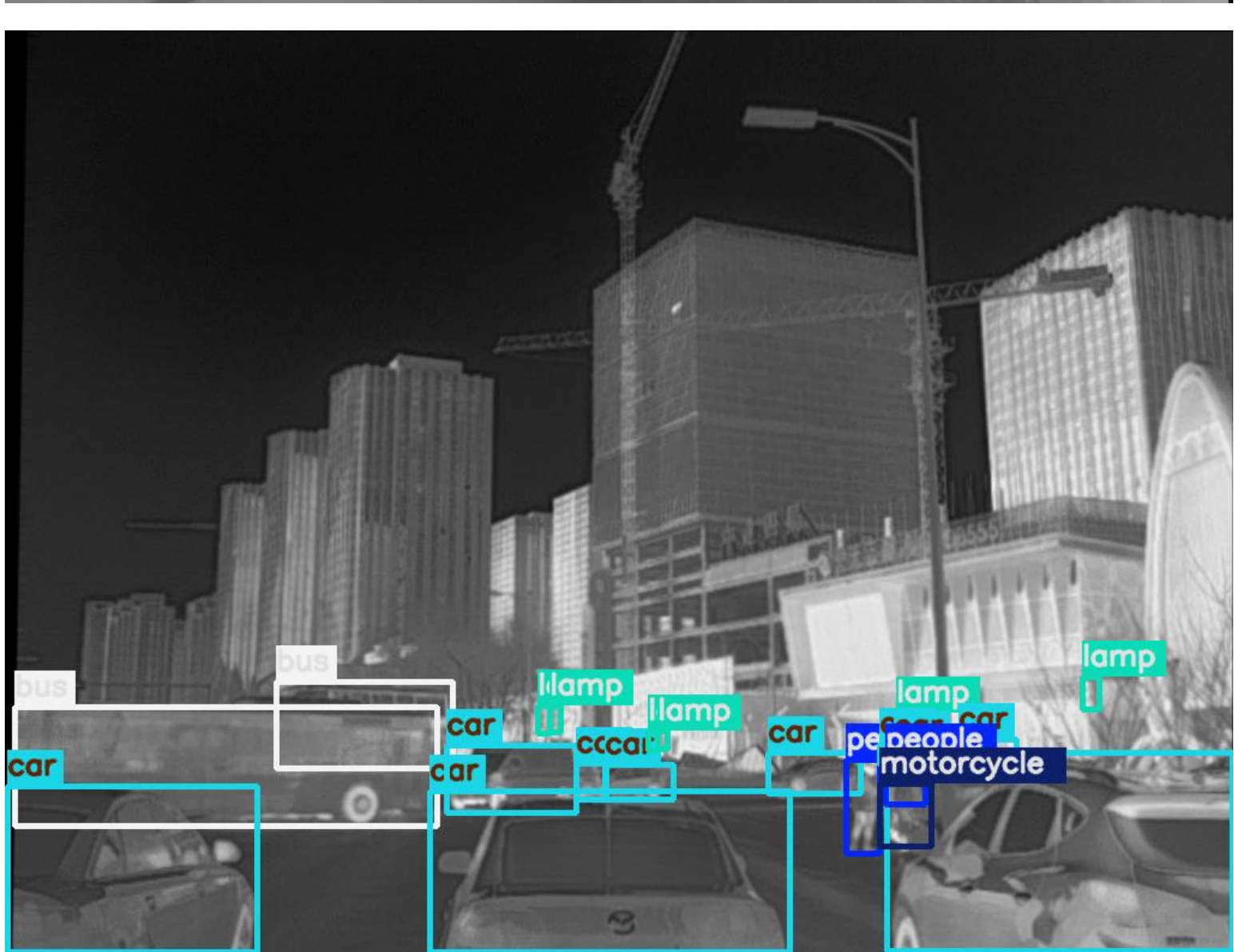
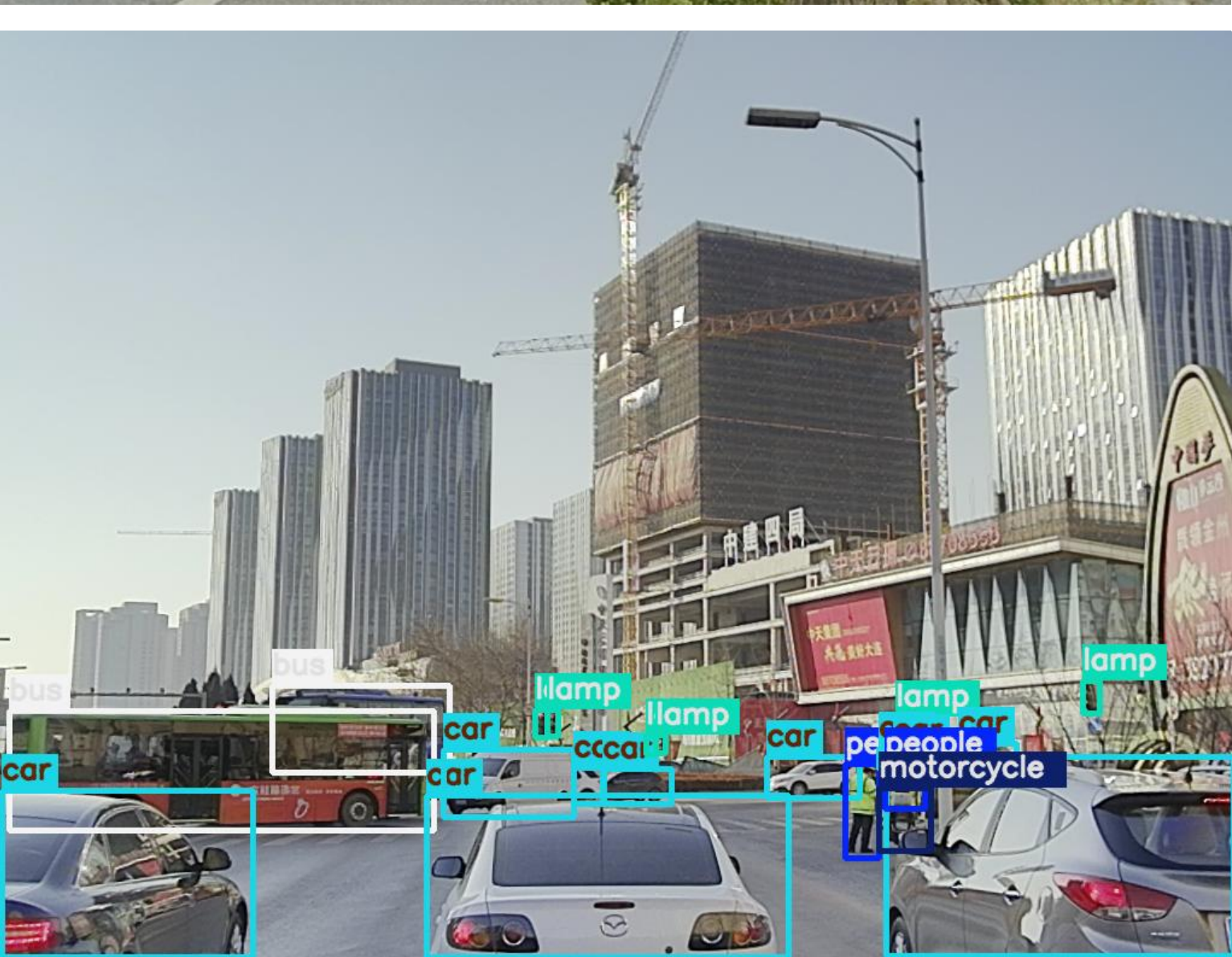
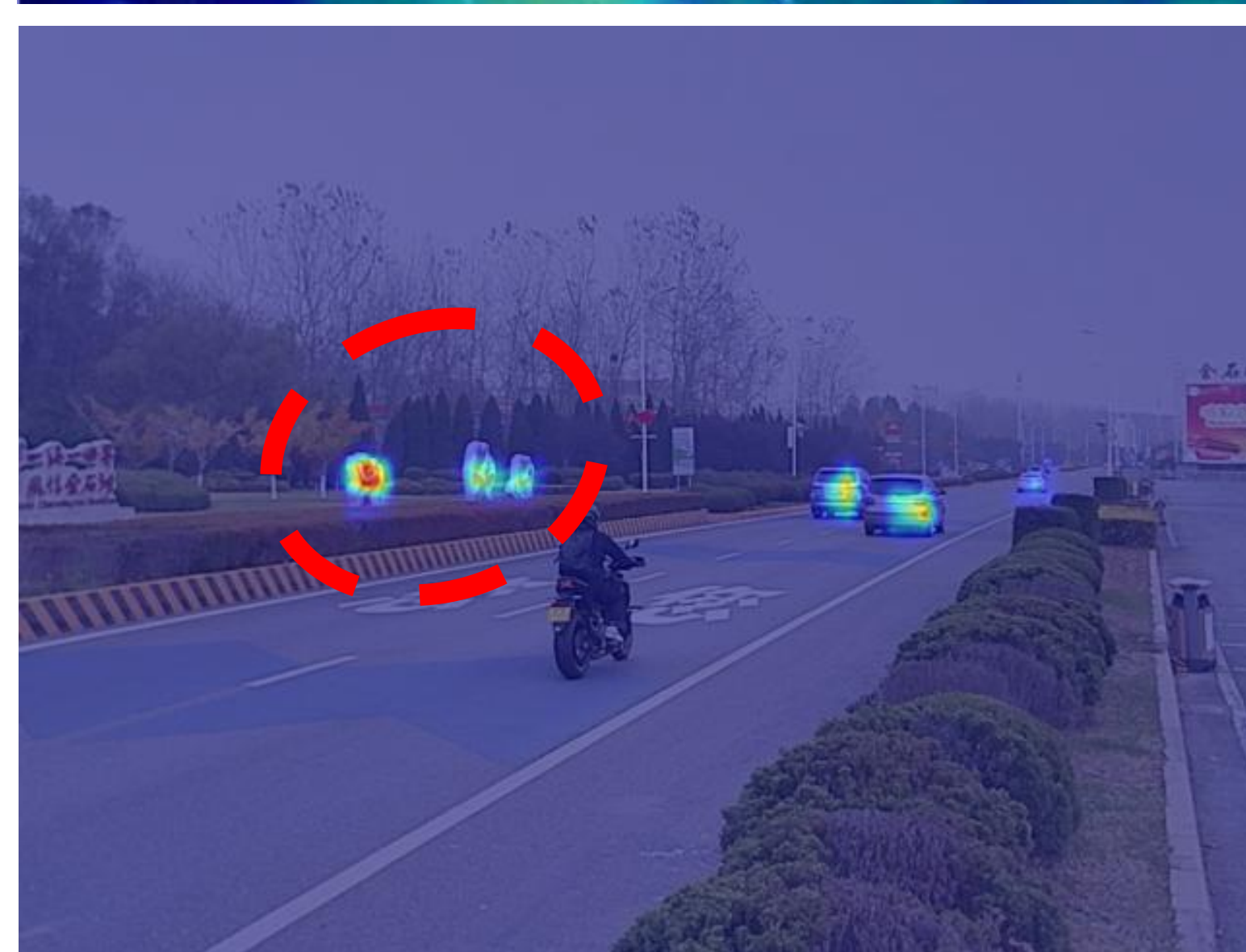
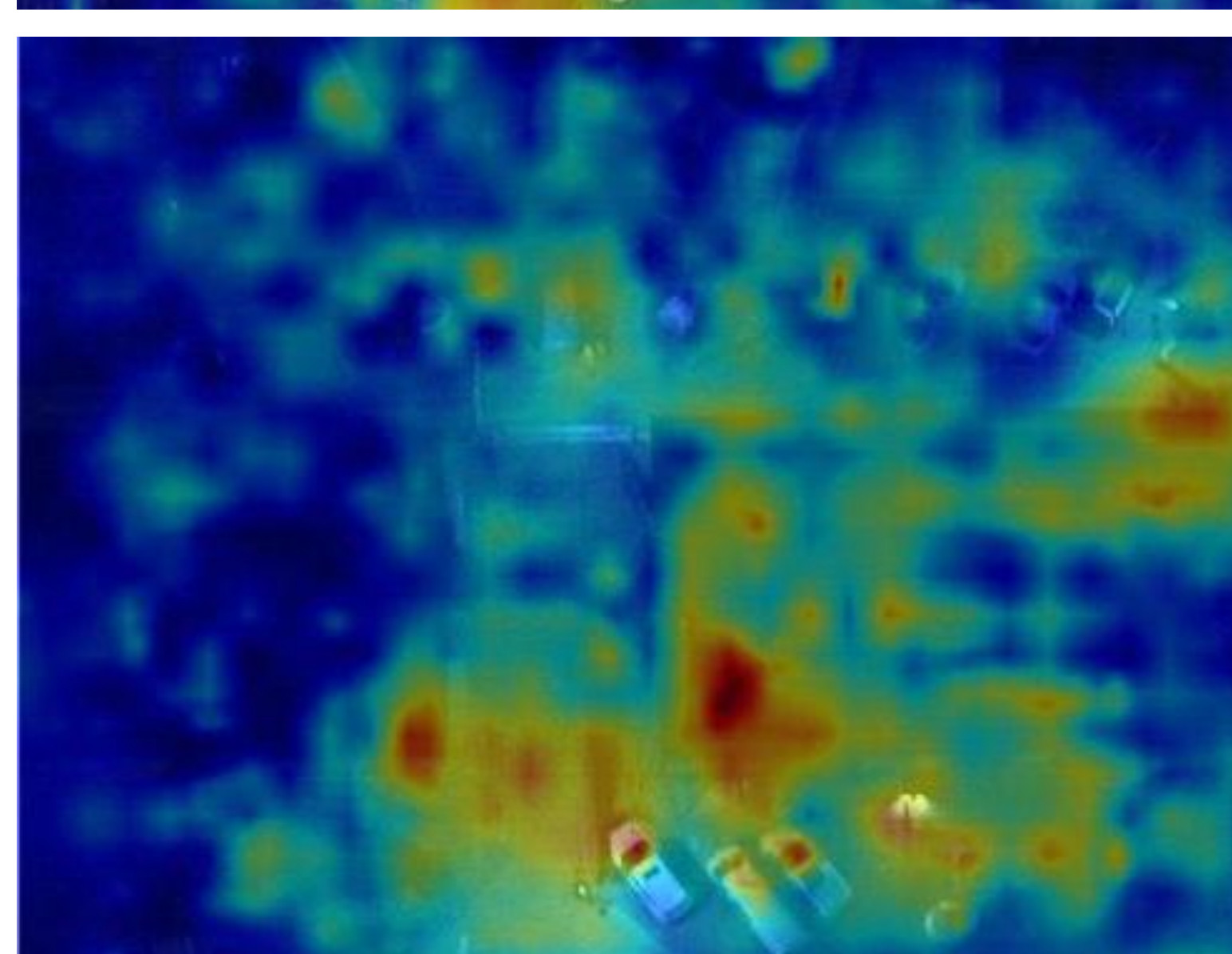
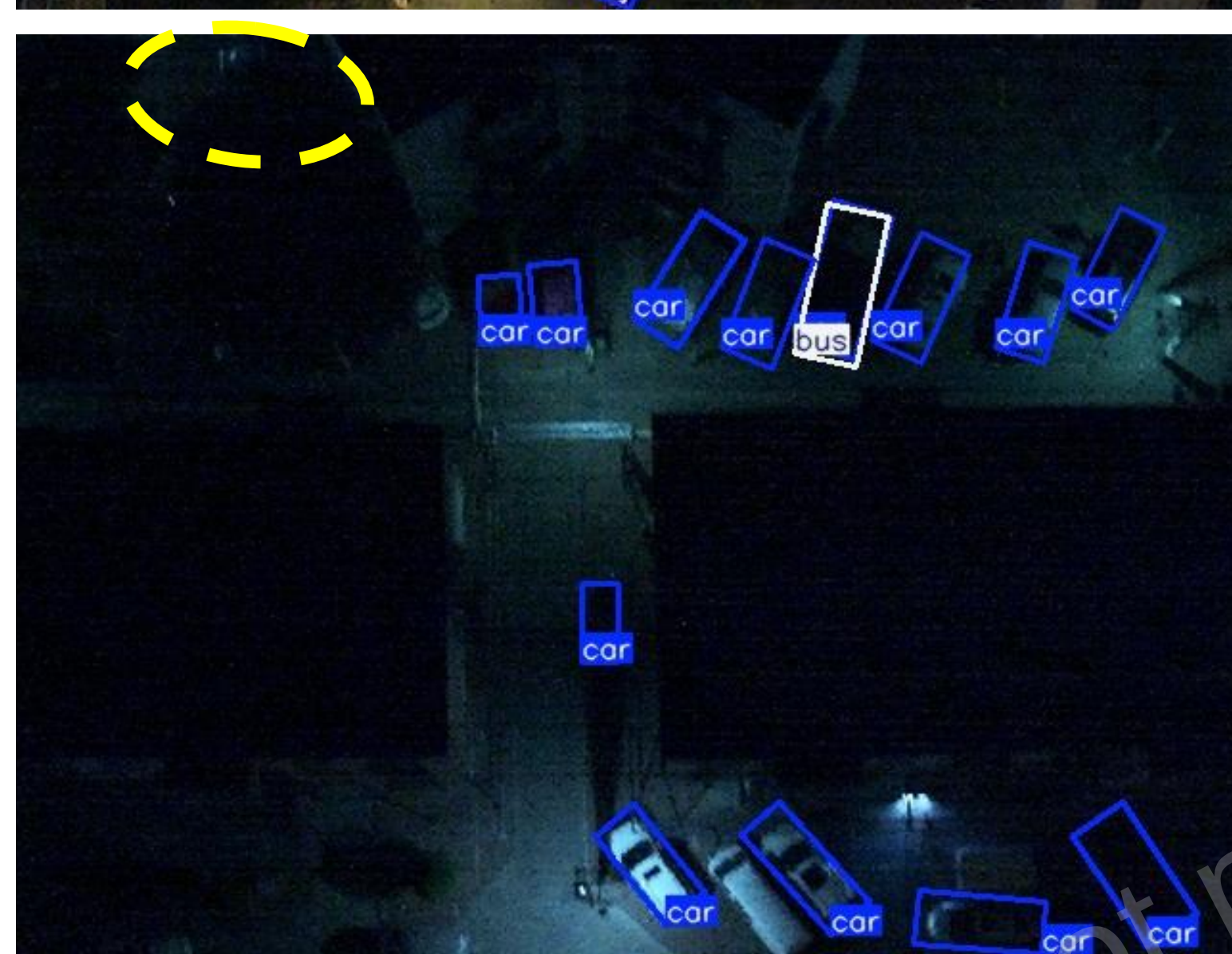
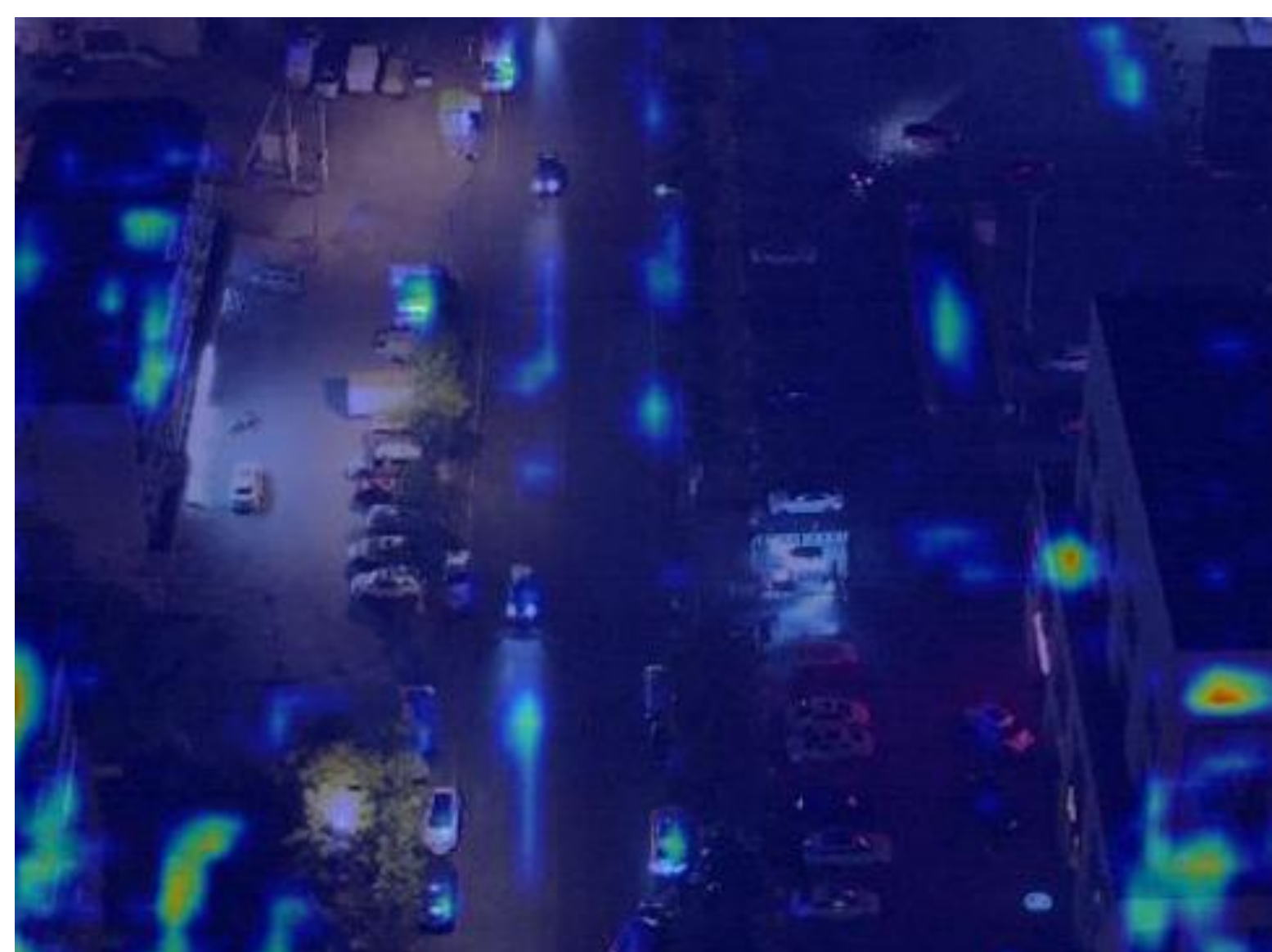
GRID



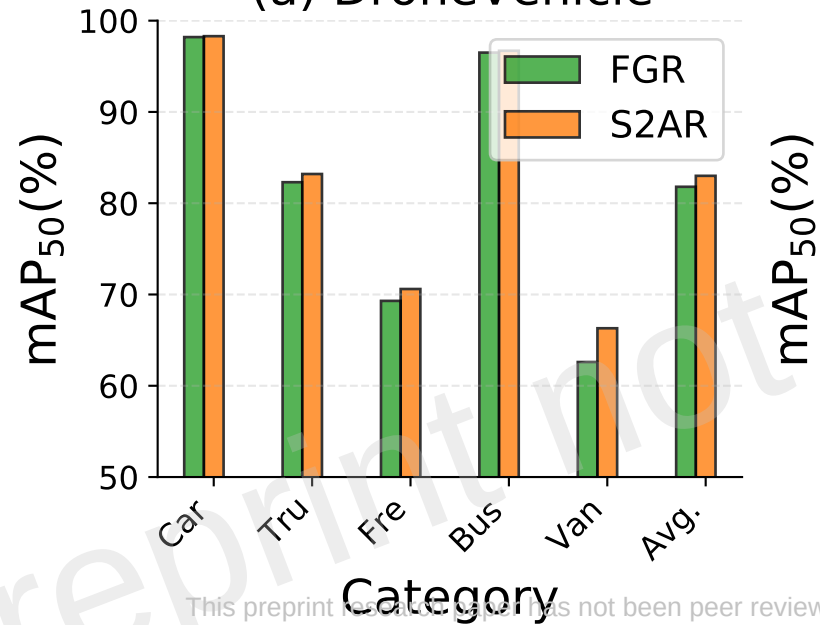
Baseline



GRID

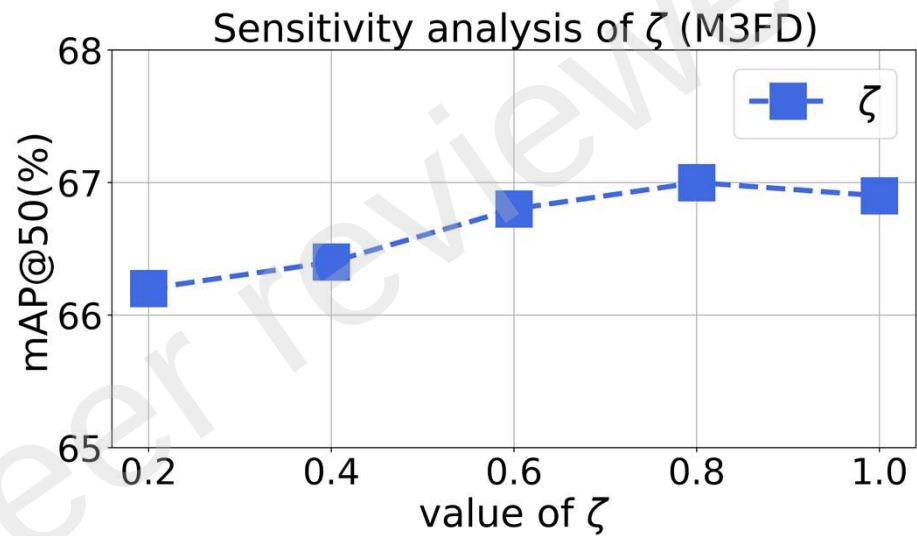
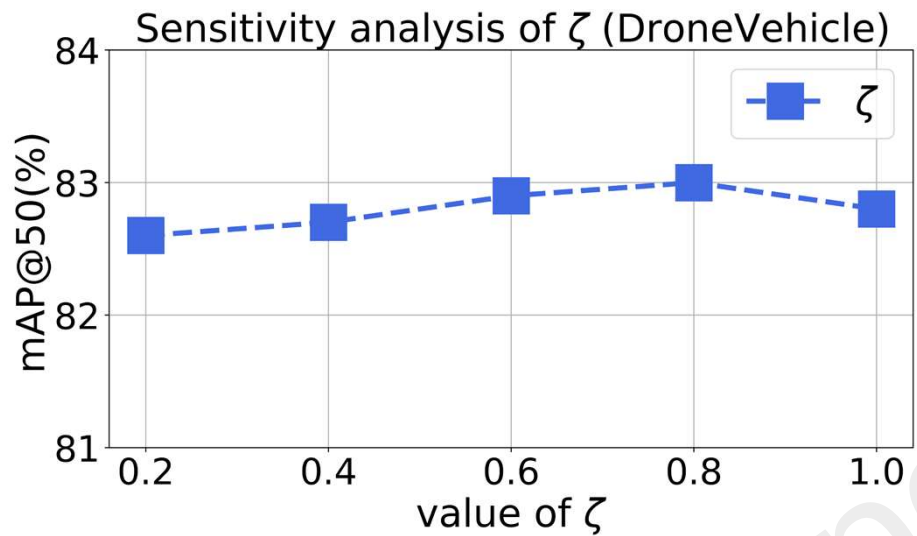


(a) DroneVehicle



(b) M3FD



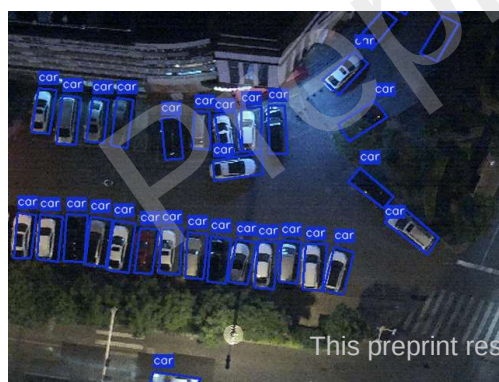
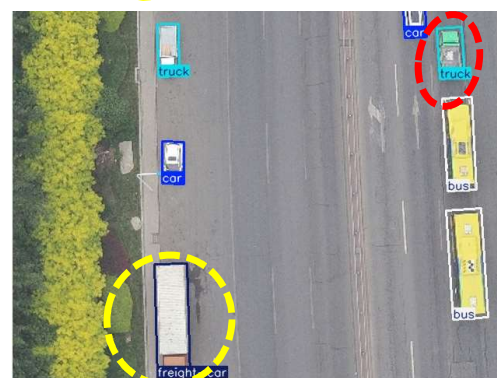
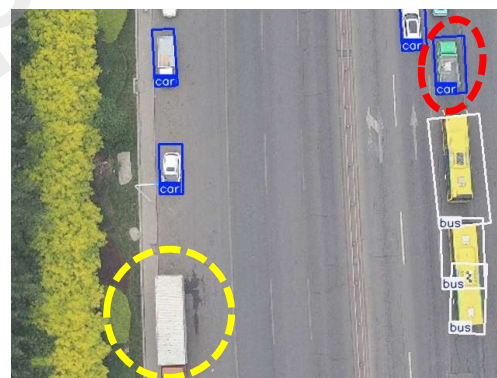
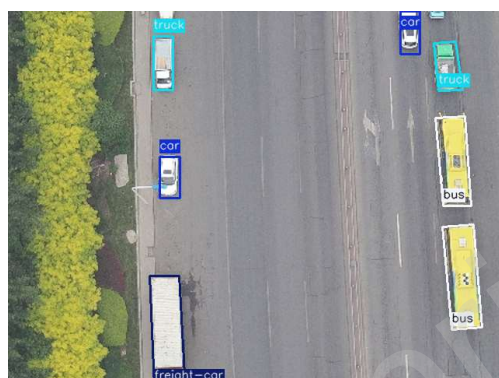
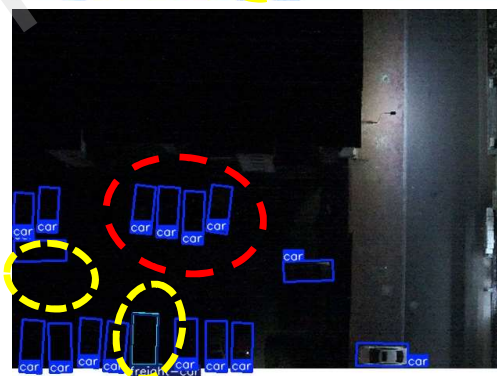
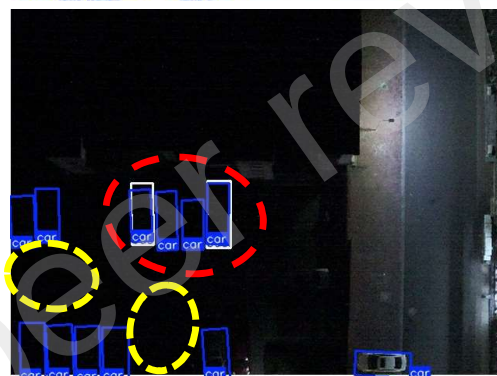
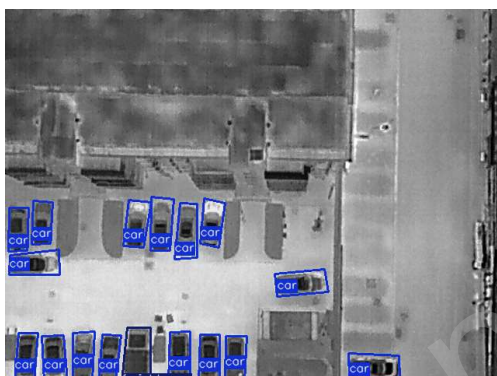
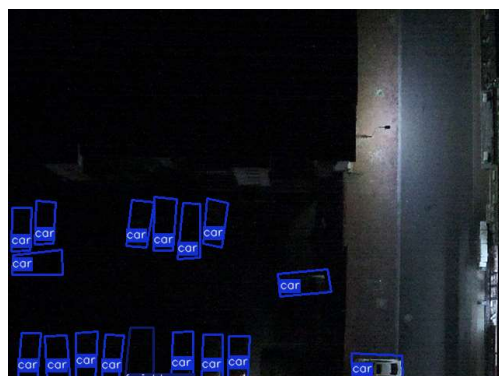
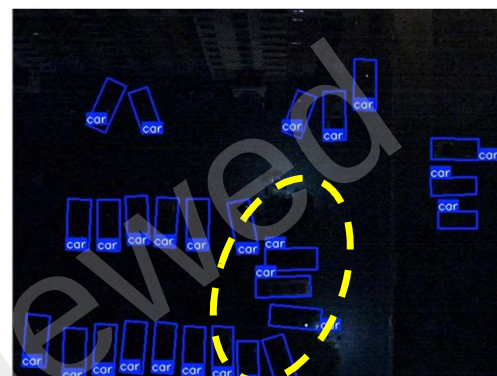
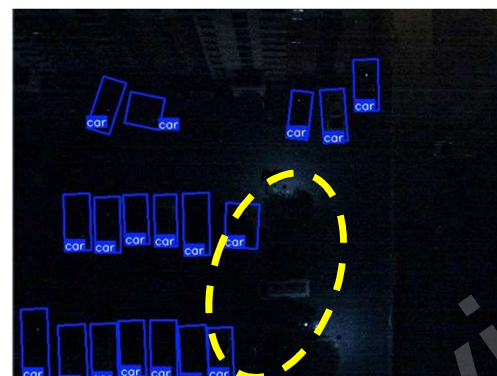
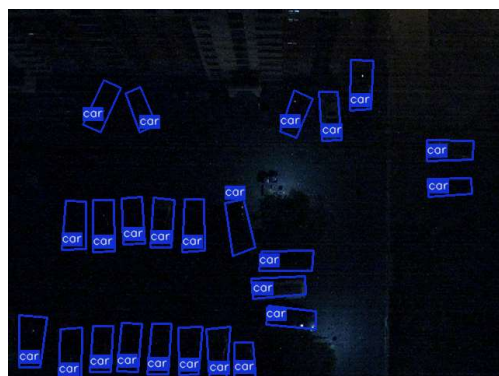


RGB Ground Truth

IR Ground Truth

Baseline

GRID

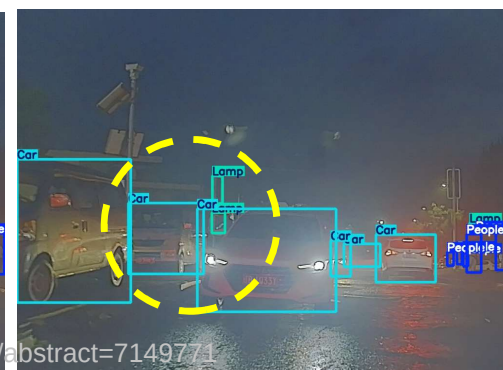
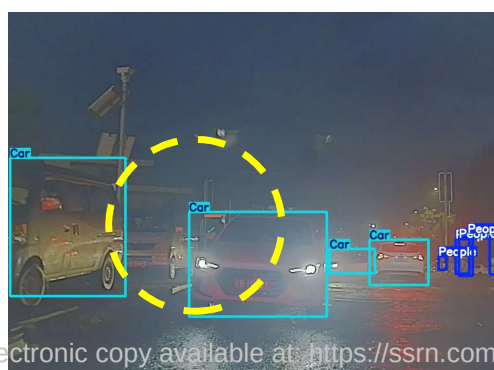
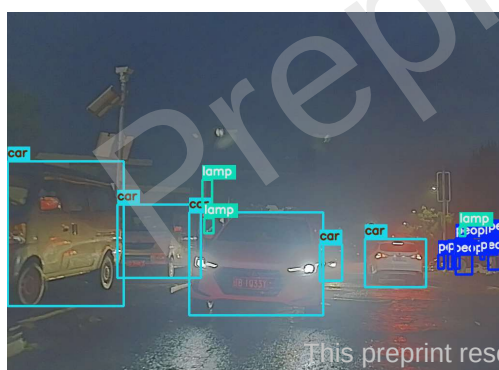
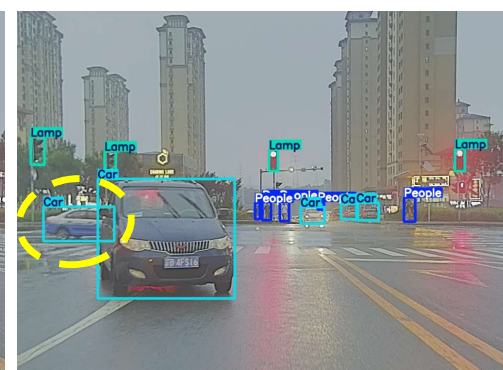
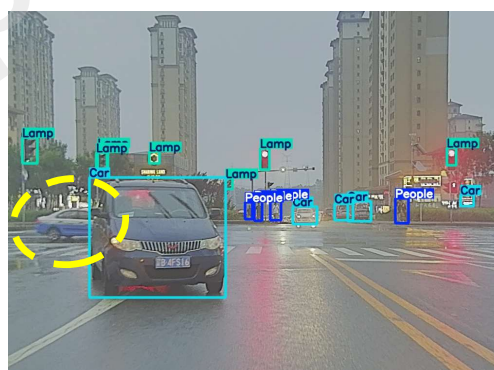
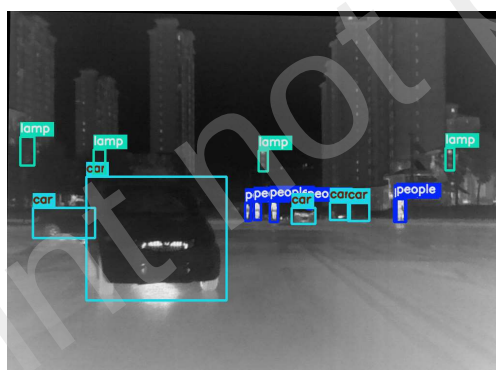
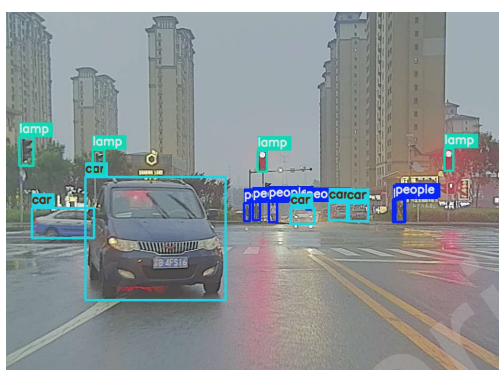
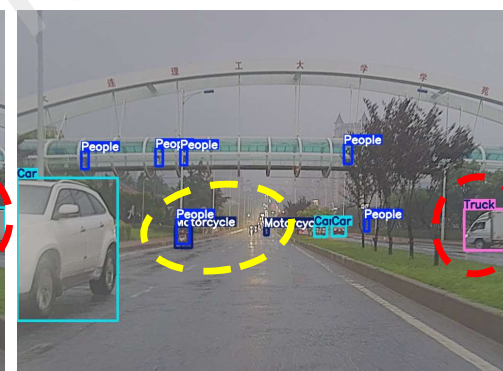
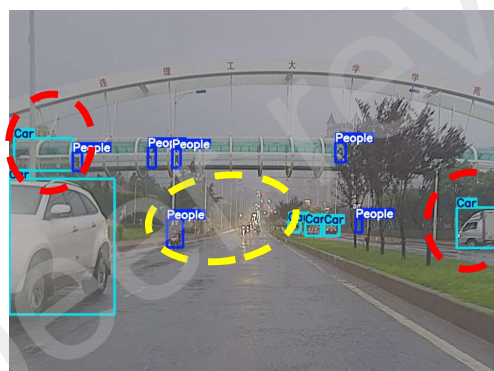
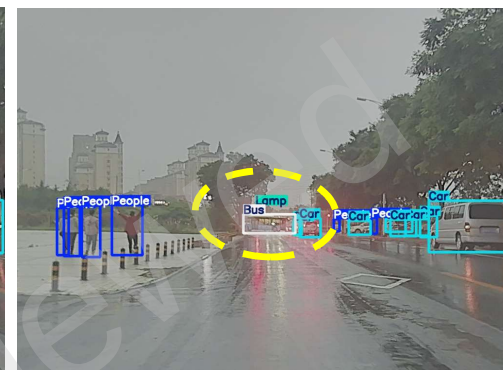
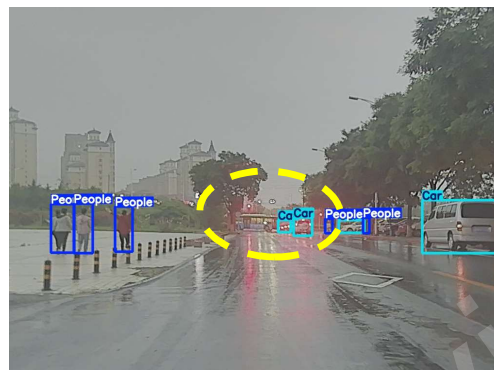
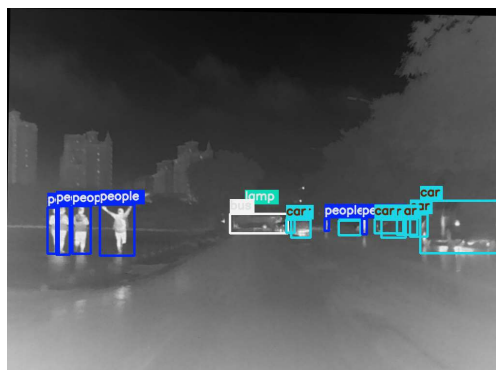


RGB Ground Truth

IR Ground Truth

Baseline

GRID



RGB Ground Truth

IR Ground Truth

Baseline

GRID

